Location-aware Channel Prediction for Efficient Digital Radio Twin

by

Subham Saha

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ABSTRACT

Digital twin (DT) is crucial for accurately modeling and simulating wireless propagation environments in next-generation (xG) networks. However, its use in real-time channel prediction remains limited due to the high computational cost of ray tracing (RT)–based emulations. This thesis introduces the *digital radio twin (DRT)*, a lightweight and real-time implementation of a DT designed to predict channel state information (CSI) efficiently. The core idea of the DRT lies in exploiting the coherence of specular reflections, a property ensuring that small variations in user equipment (UE) positions cause continuous shifts in reflection points along smooth surfaces. Building on this property, we develop a propagation path estimation (PPE)-based DRT framework that updates CSI by perturbing previously computed reflection paths rather than performing full RT. We propose two approaches to PPE: (1) manifold optimization-based; and (2) perturbation-based PPE. Extensive emulations demonstrate that the proposed method achieves lower normalized mean square error (NMSE) and reduced runtime compared to conventional methods, including dynamic RT. The proposed DRT maintains accuracy across diverse geometries and frequency bands, paving the way for intelligent and proactive wireless network operation.

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Chapter 1

Introduction

NEXT-GENERATION (xG) wireless networks, together with emerging paradigms including situation awareness [1], [2], [3], artificial intelligence (AI)-native radio access network (RAN), integrated sensing and communications (ISAC), and semantic communications, are anticipated to revolutionize sustainability, robustness, and latency. One fundamental prerequisite for achieving these advances is the acquisition of accurate CSI [4]. Traditional communication systems rely on pilot signals for the CSI acquisition [5]. However, pilot signaling introduces unnecessary delay in transmission and feedback operations, along with additional power consumption and resource overhead that scale with the number of antennas. To overcome the limitations of pilot signal-based CSI acquisition, two main approaches are widely adopted: 1) radio map-based approaches [6], [7]; and 2) DT-based approaches [8], [9]. First, in the radio map-based CSI acquisition methods, channel is characterized as a function of the UE position and the surrounding environment that enables direct CSI inference from UE position. Early studies on radio map-based CSI acquisition exploit spatial regression techniques, where channel measurements at a limited set of sample locations were obtained through measurement campaigns, and the CSI over the remaining area was inferred via interpolation [10]. While spatial regression-based radio map schemes are conceptually straightforward, they suffer from limited scalability due to the high dimensionality of CSI, which makes them impractical for large-scale or complex environments.

Recent works have introduced learning-based radio maps that exploit AI/machine learning (ML) techniques such as Gaussian mixture models (GMMs), convolutional neural networks (CNNs), and autoregressive models to capture the latent structure of CSI [11]. Compared with spatial regression methods, these AI/ML-driven learning-based maps provide more accurate CSI inference while offering better generalization and scalability across locations. Ideally, well-trained radio maps enable deterministic CSI estimation. In practice, however, their strong dependence on training data limits their applicability to the specific environment in which the measurements are collected. Consequently, whenever the environment changes, radio maps must be retrained, which requires extensive channel measurement campaigns and retraining that incur substantial cost and delay. To mitigate these limitations, several data augmentation methods have been proposed [12], [13], [14]. These methods generates synthetic training data from a small amount of measurements using generative AI models such as variational autoencoders (VAEs), generative adversarial networks (GANs), and diffusion models. While data augmentation methods partially alleviate the training dataset acquisition

overhead, they share the same limitation as learning-based radio maps where any change in the environment still necessitates retraining from scratch.

Second, in the DT-based CSI acquisition methods, CSI is obtained by emulating wireless signal propagation through RT. DTs construct high-fidelity wireless environments that replicate physical objects, materials, and mobility patterns. In this setting, RT emits a large number of candidate rays from the transmitter and traces their trajectories as they interact with surrounding structures. By aggregating the contributions of all significant rays at the receiver, DTs can reconstruct the channel with high accuracy. Unlike radio map-based methods, which rely on environment-specific training data, DTs are grounded in physical modeling and can therefore accommodate environmental changes by updating the digital replica rather than retraining the channel inference model from scratch. One major drawback of the DT-based methods is the substantial RT overhead required for high-fidelity wireless environment modeling [15]. In particular, tracing reflections, diffractions, and scatterings, together with geometric calculations, frequency-dependent modeling, and aggregation of multipath contributions across receivers incurs high computational complexity. This substantially increases both processing time and memory consumption, especially in dense urban or complex indoor environments where a large number of rays is required to achieve high-resolution accuracy, thereby hindering the adoption of DTs in real-time applications [15].

The fundamental question related to efficient wireless channel acquisition is how to obtain accurate channel information in DTs while reducing reliance on computationally intensive RT. The answer to this question will expand the applicability of DTs in emerging real-time applications such as autonomous vehicles, aerial platforms, and immersive extended reality (XR), paving the way for scalable and accurate channel acquisition in dynamic environments.

This thesis aims to develop a *digital radio twin* (DRT), an efficient DT framework tailored for real-time wireless propagation simulation. The core operation of the proposed DRT framework is the geometry-driven PPE, which predicts the multipath characteristics at nearby UE locations from those at the current location by exploiting the geometric consistency of propagation paths. The proposed PPE operation supports direct inference of CSI at adjacent UE positions, thereby substantially reducing the frequency of RT and the associated latency and power consumption. The key challenge in implementing PPE is to determine a chain of reflection points on the scatterer surfaces that satisfy the law of reflection (also known as Fermat's principle), which is formulated as a constrained optimization problem.

To address this challenge, we introduce two PPE methods: 1) a manifold optimization-based method and 2) a perturbation-based method. First, the manifold optimization-based PPE method leverages the property that object surfaces in DTs can be decomposed into smooth surface patches, such as triangles and squares, on which the notion of differentiability can be defined. Building on the differentiable structure of scatterer surfaces, we transform the constrained PPE problem into an unconstrained problem on the Riemannian manifold of scatterer surfaces and then solve it using the Riemannian conjugate gradient descent (RCG) method, thereby achieving accurate CSI prediction. Second, the perturbation-based PPE method solves the PPE problem by deriving the first-order Taylor expansions of the signal propagation chains from its Karush-Kuhn-Tucker (KKT) conditions. This approach implicitly handles the surface constraints through Lagrangian formulation, thereby reducing

the complexity of PPE. The key contributions of this thesis are summarized as follows:

- we introduce an efficient DRT framework that replaces exhaustive RT with geometry-driven signal propagation prediction;
- we develop an efficient PPE technique that characterizes local variations in signal propagation chains via manifold optimization and perturbation analysis; and
- we demonstrate the channel prediction accuracy of the proposed DRT framework through extensive simulations on the NVIDIA™Sionna platform.

Chapter 2

System Model

2.0.1 Ray Tracing-based Channel Model

We consider an upper mid-band (also known as frequency range (FR) 3 band) multiple-input multiple-output (MIMO) system where a BS equipped with $M^{\mathrm{BS}} = M_{\mathrm{h}}^{\mathrm{BS}} \times M_{\mathrm{v}}^{\mathrm{BS}}$ uniform rectangular array (URA) antennas serves a UE equipped with $M^{\mathrm{UE}} = M_{\mathrm{h}}^{\mathrm{UE}} \times M_{\mathrm{v}}^{\mathrm{UE}}$ URA antennas, within a physical environment denoted by $\mathcal{E}$. We employ a global coordinate system (GCS) where the positions of BS and UE are denoted as $\boldsymbol{x}^{\mathrm{BS}}$ and $\boldsymbol{x}^{\mathrm{UE}}$, respectively. In the upper mid-band, due to strong directivity and high path loss, the wireless channel exhibits sparse scattering characteristics, with most of the signal power concentrated in the line of sight (LOS) path and a few non-line of sight (NLOS) paths. Among the NLOS paths, the dominant components typically arise from specular reflections and diffractions [16]. Based on this property, we adopt a RT-based channel model consisting of L dominant paths. Specifically, the ray $\boldsymbol{s}_l \in \mathbb{R}^{3N_l}$ is defined as a concatenation of the ordered sequence of interaction points (specular reflection or diffraction) that forms the lth path:

$$\boldsymbol{s}_l = \left[\left(\boldsymbol{x}_l^{(1)}\right)^{\mathrm{T}} \left(\boldsymbol{x}_l^{(2)}\right)^{\mathrm{T}} \cdots \left(\boldsymbol{x}_l^{(N_l)}\right)^{\mathrm{T}} \right]^{\mathrm{T}}. \tag{2.1}$$

where N_l is the number of bounces in the signal propagation chain and $\boldsymbol{x}_l^{(i)}$ is the interaction point at the ith bounce. The interaction chain originates at the BS, i.e., $\boldsymbol{x}_l^{(0)} = \boldsymbol{x}^{\mathrm{BS}}$ and terminates at the UE, i.e., $\boldsymbol{x}_l^{(N_l+1)} = \boldsymbol{x}^{\mathrm{UE}}$. Also, the set of all interaction chains for the UE position $\boldsymbol{x}^{\mathrm{UE}}$ is denoted by

$$\mathcal{S}(\boldsymbol{x}^{\mathrm{UE}}) \triangleq \{\boldsymbol{s}_l \,:\, l = 1, 2, \ldots, L\}. \tag{2.2}$$

As the UE moves from $\boldsymbol{x}^{\mathrm{UE}}$ to $\check{\boldsymbol{x}}^{\mathrm{UE}}$, the corresponding set of interaction chains is updated to $\mathcal{S}(\check{\boldsymbol{x}}^{\mathrm{UE}})$. Once the interaction chains are determined, the electro-magnetic (EM) field solver computes the channel parameters of each propagation path, including the angle of departure (AOD), angle of arrival (AOA), channel gain, and delay, based on the geometric configuration as well as the propagation and surface properties of the environment. Let (θ_l^{BS}, ϕ_l^{BS}), (θ_l^{UE}, ϕ_l^{UE}), α_l, and τ_l be the elevation and azimuth AODs, the azimuth and elevation AOAs, the complex channel gain, and the delay of the lth path, respectively. Then, the

collection of these channel parameters $\mathcal{C}$ for all paths is expressed as

$$\mathcal{C} \triangleq \left\{ \left(\theta_l^{\mathrm{BS}}, \phi_l^{\mathrm{BS}}, \theta_l^{\mathrm{UE}}, \phi_l^{\mathrm{UE}}, \alpha_l, \tau_l\right) \,:\, l = 1, 2, \ldots, L \right\} \tag{2.3}$$

$$= f_{\mathrm{EM}}\big(\, \mathcal{S}(\boldsymbol{x}^{\mathrm{UE}});\, \mathcal{E}\,\big)\,. \tag{2.4}$$

We now describe the detailed procedure of the EM solver. Let $\boldsymbol{k}_l^{\mathrm{BS}} \in \mathbb{R}^3$ and $\boldsymbol{k}_l^{\mathrm{UE}} \in \mathbb{R}^3$ be the BS and UE direction vectors of the lth path, respectively. The BS direction vector is computed based on the first interaction point and the BS position, while the UE direction vector is computed based on the UE position and the last interaction point:

$$\boldsymbol{k}_l^{\mathrm{BS}} = \frac{\boldsymbol{x}_l^{(1)} - \boldsymbol{x}^{\mathrm{BS}}}{\left\|\boldsymbol{x}_l^{(1)} - \boldsymbol{x}^{\mathrm{BS}}\right\|_2}, \quad \boldsymbol{k}_l^{\mathrm{UE}} = \frac{\boldsymbol{x}_l^{(N_l)} - \boldsymbol{x}^{\mathrm{UE}}}{\left\|\boldsymbol{x}_l^{(N_l)} - \boldsymbol{x}^{\mathrm{UE}}\right\|_2}\,. \tag{2.5}$$

The relationship between the direction vectors and the azimuth and elevation angles is given by

$$\begin{aligned} \boldsymbol{k}_l^{\mathrm{BS}} &\triangleq \begin{bmatrix} \sin\theta_l^{\mathrm{BS}}\cos\phi_l^{\mathrm{BS}} & \sin\theta_l^{\mathrm{BS}}\sin\phi_l^{\mathrm{BS}} & \cos\theta_l^{\mathrm{BS}} \end{bmatrix}^{\mathrm{T}} \\ \boldsymbol{k}_l^{\mathrm{UE}} &\triangleq \begin{bmatrix} \sin\theta_l^{\mathrm{UE}}\cos\phi_l^{\mathrm{UE}} & \sin\theta_l^{\mathrm{UE}}\sin\phi_l^{\mathrm{UE}} & \cos\theta_l^{\mathrm{UE}} \end{bmatrix}^{\mathrm{T}} \end{aligned} \tag{2.6}$$

Using (2.6), the azimuth and elevation AOAs and AODs can be expressed as

$$\theta_l^{\mathrm{BS}} = \arccos([\,\boldsymbol{k}_l^{\mathrm{BS}}\,]_3), \qquad \phi_l^{\mathrm{BS}} = \mathrm{atan2}([\,\boldsymbol{k}_l^{\mathrm{BS}}\,]_2, [\,\boldsymbol{k}_l^{\mathrm{BS}}\,]_1) \tag{2.7}$$

$$\theta_l^{\mathrm{UE}} = \arccos([\,\boldsymbol{k}_l^{\mathrm{UE}}\,]_3), \qquad \phi_l^{\mathrm{UE}} = \mathrm{atan2}([\,\boldsymbol{k}_l^{\mathrm{UE}}\,]_2, [\,\boldsymbol{k}_l^{\mathrm{UE}}\,]_1)\,. \tag{2.8}$$

Furthermore, let $d_l^{(i)} \triangleq \left\|\boldsymbol{x}_l^{(i)} - \boldsymbol{x}_l^{(i-1)}\right\|_2$ be the distance between two successive interaction points $\boldsymbol{x}_l^{(i-1)}$ and $\boldsymbol{x}_l^{(i)}$. Then, the propagation delay of the lth path is given by

$$\tau_l = \frac{\sum_{i=1}^{N_l+1} d_l^{(i)}}{c} \tag{2.9}$$

where c denotes the speed of light.

Next, we explain the path gain computation including the free-space path loss, and transformations due to specular reflection and diffraction. The electric fields associated with each antenna element at the BS and UE are assumed to be uniform. The incident and outgoing fields at the ith interaction point $\boldsymbol{x}_l^{(i)}$ are denoted by $\boldsymbol{e}_{l,\,\mathrm{IN}}^{(i)}$ and $\boldsymbol{e}_{l,\,\mathrm{OUT}}^{(i)}$, respectively. Between the $(i-1)$th and ith interaction points, the field undergoes a transformation due to free-space path loss as

$$\boldsymbol{e}_{l,\,\mathrm{IN}}^{(i)} = \frac{1}{d_l^{(i)}} \boldsymbol{e}_{l,\,\mathrm{OUT}}^{(i-1)}\,. \tag{2.10}$$

In case of a specular reflection, the electric field undergoes another transformation given by

$$\boldsymbol{W}_l^i\, \boldsymbol{e}_{l,\,\mathrm{OUT}}^{(i)} = \begin{bmatrix} r_{\perp,l}^{(i)} & 0 \\ 0 & r_{\parallel,l}^{(i)} \end{bmatrix} \boldsymbol{W}_l^i\, \boldsymbol{e}_{l,\,\mathrm{IN}}^{(i)} \tag{2.11}$$

where $\boldsymbol{W}_l^i$ is the change-of-basis matrix from the GCS to a local orthonormal frame defined at $\boldsymbol{x}_l^{(i)}$ on the reflecting surface, and $r_{\perp,l}^{(i)}$ and $r_{\parallel,l}^{(i)}$ denote the Fresnel reflection coefficients for

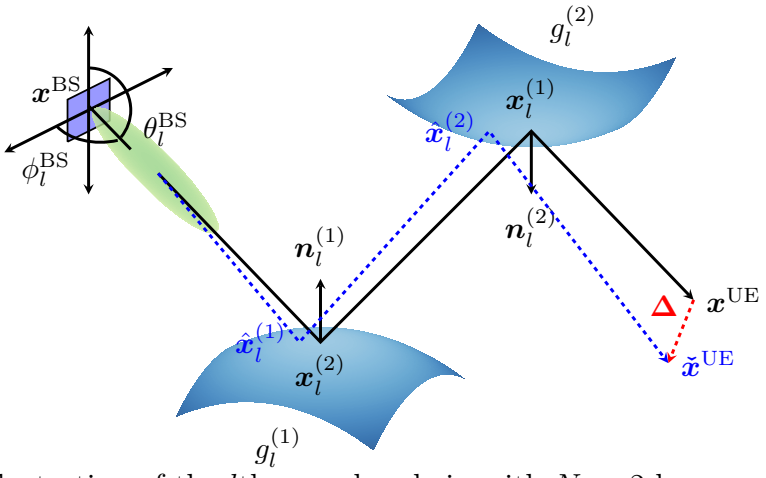


Figure 2.1: An illustration of the lth specular chain with $N_l = 2$ bounces. Reflection points $\boldsymbol{x}_l^{(1)}$ and $\boldsymbol{x}_l^{(2)}$ lie on the surfaces $g_l^{(1)}$ and $g_l^{(2)}$, respectively.

the transverse electric and magnetic fields, respectively:

$$r_{\perp,l}^{(i)} = \frac{\cos\nu_l^{(i)} - \sqrt{\left(\eta_l^{(i)}\right)^2 - \sin^2\nu_l^{(i)}}}{\cos\nu_l^{(i)} + \sqrt{\left(\eta_l^{(i)}\right)^2 - \sin^2\nu_l^{(i)}}} \tag{2.12}$$

$$r_{\parallel,l}^{(i)} = \frac{\eta_l^{(i)}\cos\nu_l^{(i)} - \sqrt{\left(\eta_l^{(i)}\right)^2 - \sin^2\nu_l^{(i)}}}{\eta_l^{(i)}\cos\nu_l^{(i)} + \sqrt{\left(\eta_l^{(i)}\right)^2 - \sin^2\nu_l^{(i)}}}\,. \tag{2.13}$$

Here, $\nu_l^{(i)}$ is the angle of incidence defined as $\cos\nu_l^{(i)} = -\left(\boldsymbol{n}_l^{(i)}(\boldsymbol{x}_l^{(i)})\right)^{\mathrm{T}} \frac{\boldsymbol{x}_l^{(i)} - \boldsymbol{x}_l^{(i-1)}}{d_l^{(i)}}$ where $\boldsymbol{n}_l^{(i)} : \mathbb{R}^3 \to \mathbb{R}^3$ is the surface normal function of the ith reflecting surface given by $\boldsymbol{n}_l^{(i)}(\boldsymbol{x}) \triangleq \frac{\nabla_{\boldsymbol{x}} g_l^{(i)}}{\|\nabla_{\boldsymbol{x}} g_l^{(i)}\|_2}$. Also, $\eta_l^{(i)}$ is the relative permittivity of the reflecting surface, which is determined by the surface conductivity and the operating frequency. In the case of diffraction, the electric field undergoes a similar transformation. However, the corresponding transformation matrix is more complex, as it depends on the properties of the two intersecting surfaces that form the edge where diffraction occurs:

$$\boldsymbol{W}_l^i\, \boldsymbol{e}_{l,\mathrm{OUT}}^{(i)} = \boldsymbol{Q}_l^{(i)}\, \boldsymbol{W}_l^i\, \boldsymbol{e}_{l,\mathrm{IN}}^{(i)}\,. \tag{2.14}$$

Here, $\boldsymbol{Q}_l^{(i)}$ denotes the interaction matrix that transforms the electric field due to diffraction at the ith interaction point in the lth path [17]. After consecutive transformations over N_l interactions, the channel gain of the lth path at the UE is calculated as

$$\alpha_l = \frac{\left(\boldsymbol{e}_{l,\mathrm{OUT}}^{(0)}\right)^{\mathrm{T}} \boldsymbol{e}_{l,\mathrm{IN}}^{(N_l+1)}}{\|\boldsymbol{e}_{l,\mathrm{OUT}}^{(0)}\|_2}\,. \tag{2.15}$$

Finally, the downlink channel frequency response (CFR) matrix $\boldsymbol{H}_{\mathcal{C}}(f) \in \mathbb{C}^{M^{\mathrm{BS}} \times M^{\mathrm{UE}}}$ is constructed using $\mathcal{C}$ as

$$\boldsymbol{H}_{\mathcal{C}}(f) = \sum_{l=1}^{L} \alpha_l e^{-j2\pi f \tau_l} \, \boldsymbol{a}_l^{\mathrm{BS}}(\theta_l^{\mathrm{BS}}, \phi_l^{\mathrm{BS}}) \left(\boldsymbol{a}^{\mathrm{UE}}(\theta_l^{\mathrm{UE}}, \phi_l^{\mathrm{UE}})\right)^{\mathrm{T}} \tag{2.16}$$

where $\boldsymbol{a}^{\mathrm{BS}}(\theta_l^{\mathrm{BS}}, \phi_l^{\mathrm{BS}})$ and $\boldsymbol{a}^{\mathrm{UE}}(\theta_l^{\mathrm{UE}}, \phi_l^{\mathrm{UE}})$ denote the BS and UE array steering vectors, respectively, defined as

$$\boldsymbol{a}^{\mathrm{BS}}(\theta_l^{\mathrm{BS}}, \phi_l^{\mathrm{BS}}) \triangleq \left[e^{j\frac{2\pi}{\lambda}(\boldsymbol{p}_1^{\mathrm{BS}})^{\mathrm{T}}\boldsymbol{k}_l^{\mathrm{BS}}} \quad e^{j\frac{2\pi}{\lambda}(\boldsymbol{p}_2^{\mathrm{BS}})^{\mathrm{T}}\boldsymbol{k}_l^{\mathrm{BS}}} \quad \cdots e^{j\frac{2\pi}{\lambda}(\boldsymbol{p}_{M^{\mathrm{BS}}}^{\mathrm{BS}})^{\mathrm{T}}\boldsymbol{k}_l^{\mathrm{BS}}} \right]^{\mathrm{T}} \tag{2.17}$$

$$\boldsymbol{a}^{\mathrm{UE}}(\theta_l^{\mathrm{UE}}, \phi_l^{\mathrm{UE}}) \triangleq \left[e^{j\frac{2\pi}{\lambda}(\boldsymbol{p}_1^{\mathrm{UE}})^{\mathrm{T}}\boldsymbol{k}_l^{\mathrm{UE}}} \quad e^{j\frac{2\pi}{\lambda}(\boldsymbol{p}_2^{\mathrm{UE}})^{\mathrm{T}}\boldsymbol{k}_l^{\mathrm{UE}}} \quad \cdots e^{j\frac{2\pi}{\lambda}(\boldsymbol{p}_{M^{\mathrm{UE}}}^{\mathrm{UE}})^{\mathrm{T}}\boldsymbol{k}_l^{\mathrm{UE}}} \right]^{\mathrm{T}}. \tag{2.18}$$

Here, $\boldsymbol{p}_m^{\mathrm{BS}}$ and $\boldsymbol{p}_m^{\mathrm{UE}} \in \mathbb{R}^3$ are the positions of the mth BS and UE antenna elements, respectively.

2.0.2 Propagation Path Estimation Problem Formulation

From (2.3) and (2.16), one can see that the CFR $\boldsymbol{H}_{\mathcal{C}}(f)$ is derived from the set of propagation paths $\mathcal{S}(\boldsymbol{x}^{\mathrm{UE}})$, which in turn depends on the UE position $\boldsymbol{x}^{\mathrm{UE}}$ and the wireless environment $\mathcal{E}$. We assume that the UE position belongs to the set $\mathcal{U}$. While $\mathcal{E}$ is directly provided by the DT, $\mathcal{S}(\boldsymbol{x}^{\mathrm{UE}})$ must be estimated. The key principle for determining $\mathcal{S}(\boldsymbol{x}^{\mathrm{UE}})$ is to identify the sequence of interaction points that connects the BS and the UE with the shortest path length. The total path length of the lth chain is expressed as $\sum_{i=1}^{N_l+1} \|\boldsymbol{x}_l^{(i)} - \boldsymbol{x}_l^{(i-1)}\|_2$. In addition, the interaction points must lie on their corresponding surfaces, and therefore satisfy the surface constraints $g_l^{(i)}(\boldsymbol{x}_l^{(i)}) = 0$ where $g_l^{(i)} : \mathbb{R}^3 \to \mathbb{R}$ is the implicit function of the reflecting surface associated with the ith bounce of the lth path. In case of reflection, the surface is a two-dimensional object while for diffraction, the surface collapses to a one-dimensional object (e.g., a line or edge). The shortest path problem $\mathscr{P}_l$ for the lth specular chain can be formulated as follows:

$$\mathscr{P}_l: \quad \underset{\mathcal{S}_l}{\text{minimize}} \quad \sum_{i=1}^{N_l+1} \left\|\boldsymbol{x}_l^{(i)} - \boldsymbol{x}_l^{(i-1)}\right\|_2 \tag{2.19a}$$

$$\text{subject to} \quad g_l^{(i)}\left(\boldsymbol{x}_l^{(i)}\right) = 0, \quad \forall i = 1, 2, \cdots, N_l \tag{2.19b}$$

$$\boldsymbol{x}_l^{(0)} = \boldsymbol{x}^{\mathrm{BS}} \tag{2.19c}$$

$$\boldsymbol{x}_l^{(N_l+1)} = \boldsymbol{x}^{\mathrm{UE}} \tag{2.19d}$$

However, directly solving $\mathscr{P}_l$ is challenging because the surface constraints in (2.19b) are not in general affine, rendering $\mathscr{P}_l$ non-convex. Furthermore, scene geometry itself may contain nonconvex objects, such as concave curved structures or recessed building facades. This nonconvexity can destroy the uniqueness of the solution; for example, as the UE moves, a single reflection path may split into multiple paths or distinct paths may merge on the same surface. Such path multiplicity causes substantial channel prediction errors.

2.0.3 Conventional Digital Twin-based Channel Prediction Methods

Shooting and Bouncing of Rays Method

For a given transmitter-receiver pair, RT identifies the set of propagation paths that satisfy Snell's law and the edge-diffraction conditions. To do so, RT typically adopts the shooting and bouncing of rays (SBR) method that launches multiple rays from the transmitter in different directions, traces their interactions with the environment, and finds the candidate rays reaching the receiver.

While the SBR method is straightforward, it requires a large number of sampled rays to explore all possible specular paths, resulting in high computational complexity. Specifically, the computational cost of RT mainly arises from three operations: 1) launching rays; 2) computing ray-object intersections; and 3) tracing the recursive sequence of reflections, transmissions, and diffractions for each ray.[1] In practice, the maximum number of reflections that a ray can undergo is bounded, since the energy carried by the ray diminishes with each reflection due to material absorption and scattering losses. Accordingly, we impose a maximum reflection order of K. Also, let R be the number of ray directions sampled for the scene. Then, the overall computational complexity of RT is given by $RK\mathcal{O}(\log N)$ where N is the number of objects in the scene. Although K and N are finite, R can be arbitrarily large depending on the scene. Moreover, since the accuracy of the propagation path estimation depends on the number of sampled rays, achieving accurate channel acquisition generally requires substantial complexity.

Image Method

To alleviate the computational burden of the SBR method, an image method (IM) that replaces specular reflection with an equivalent image source was introduced [18]. Specifically, the IM first employs the SBR method to enumerate candidate specular chains, and then performs two subsequent steps. i.e., forward propagation and backtracking. In the forward propagation step, the images of the BS on the reflecting surfaces are computed along each candidate specular chain. In the backtracking step, the corresponding reflection points are recovered from these images and then verified to ensure that they lie on the reflecting surfaces and that the resulting chain remains unobstructed by other objects. The main limitation of the IM is its reliance on planar reflecting surfaces. For nonplanar reflectors, the virtual image source is not uniquely defined, so the backtracking stage must solve for admissible reflection points rather than recover them directly. This weakens path pruning and may allow multiple valid paths to arise from a single candidate specular chain, thereby reducing the channel acquisition accuracy.

[1] Among these operations, ray-object intersection computation is typically the most computationally expensive. To reduce computational complexity, practical SBR implementations employ several optimization techniques, including bounding volumes, hierarchical spatial partitioning methods (e.g., bounding volume hierarchy (BVH) and k-d tree), and early ray termination.

Dynamic Ray Tracing

To achieve a balance between channel acquisition accuracy and computational complexity, a technique called dynamic RT was introduced recently [8], [9]. Its key principle is the coherence of specular reflection such that for sufficiently small perturbations in the transmitter and receiver positions, the reflection points vary continuously over smooth surfaces [19]. Leveraging this property, dynamic RT uses IM to update specular chains over intervals during which the chains remain unchanged. To be specific, for each specular chain, the forward step updates the image sources of the BS according to the new UE position, after which the backtracking step predicts the updated reflection points by recomputing the ray-surface intersections.

While dynamic ray tracing offers a more efficient alternative by predicting future specular paths through iterative image construction, its efficiency still relies heavily on the assumption of planar reflecting surfaces, much like the IM. When this assumption does not hold, the image construction process becomes ambiguous, which leads to redundant candidate paths and reduces the effectiveness of pruning during backtracking. Given that common rendering software supports both planar and nonplanar geometric objects, this planar surface assumption can significantly limit the applicability of dynamic ray tracing in practical environments.

Min-Path-Tracing

While the SBR method, the IM, and dynamic ray tracing solve $\mathscr{P}_l$ indirectly, the min-path-tracing (MPT) method proposed in [20] directly formulates $\mathscr{P}_l$ as a minimization problem. In particular, MPT first converts $\mathscr{P}_l$ into an unconstrained optimization problem by augmenting the objective function with penalty terms that capture violations of physical propagation laws and geometric constraints. Then, it employs the gradient descent method to perform the minimization of the augmented objective function. The main drawbacks of MPT come from how it handles geometric constraints. MPT enforces spatial boundaries using an extrinsic penalty function defined by analytical surface equations. This formulation is limited in two key ways. First, it induces a highly non-convex loss landscape, requiring heuristic restarts when gradient descent steps drift into the ambient space. Second, it is incompatible with irregular discrete triangular meshes used in discrete manifold-based DTs.

Chapter 3

Perturbation-based Propagation Path Estimation

While manifold optimization-based PPE (PPE-MO), discussed in the next chapter, adopts the update step from manifold optimization, its primary challenge lies in handling the mesh representation of the scene. If the PPE process is offloaded to the edge, managing the mesh data structure can become a computational bottleneck, particularly for small displacements in the UE position. In such cases, the interaction points typically remain within the same mesh face and do not require information about neighboring faces. To address this, the perturbation-based PPE is proposed as a lightweight alternative to PPE-MO. It operates using only the geometric information available from the RT solution and does not rely on a complex mesh data structure. The core idea of perturbation-based PPE (PPE-PRT) is to update the interaction points using a first-order approximation of their variation with respect to the UE position. The remainder of this section presents the problem formulation for PPE-PRT and describes the corresponding update method for the interaction points.

3.1 Methodology

For the interaction chain l, the reflection points depend on the positions of the UE, the BS, and the surface functions $g_l^{(i)}$, for $i \in \{1, \cdots, N_l\}$. The path function ψ_l that provides $\boldsymbol{s}_l$ is defined as:

$$\boldsymbol{s}_l = \psi_l\big(\boldsymbol{x}^{\mathrm{BS}}, \boldsymbol{x}^{\mathrm{UE}}, g_l^{(1)}, g_l^{(2)}, \cdots, g_l^{(N_l)}\big)\,. \tag{3.1}$$

The path function is obtained as the solution of the optimization problem $\mathscr{P}$. Since the BS is fixed, and due to the coherence property of specular reflections, small changes in the UE position imply that ψ_l depends only on $\boldsymbol{x}^{\mathrm{UE}}$ and the reflection surfaces $g_l^{(i)}$. For notational convenience, let $\boldsymbol{\psi}_l^{(i)}$ denote the path function restricted to surface i. With a change in the UE position, the new reflection point can be approximated using a first-order Taylor series expansion:

$$\hat{\boldsymbol{x}}_l^{(i)} \approx \boldsymbol{x}_l^{(i)} + \frac{\partial \boldsymbol{\psi}_{l,\boldsymbol{x}}^{(i)}}{\partial \boldsymbol{x}^{\mathrm{UE}}}\,\Delta \boldsymbol{x}^{\mathrm{UE}}\,. \tag{3.2}$$

Here $\boldsymbol{\psi}_{l,\boldsymbol{x}}^{(i)}$ denotes the component of the path function responsible for $\boldsymbol{x}_l^{(i)}$. The challenge, however, is that the path function $\boldsymbol{\psi}_l^{(i)}$ does not have an explicit form, making it impossible to directly compute its derivatives. Nevertheless, the first-order derivatives of $\boldsymbol{\psi}_l^{(i)}$ can be derived from the Lagrangian formulation of $\mathscr{P}$. The perturbation-based PPE utilizes the first-order derivatives of ψ_l to estimate $\boldsymbol{s}_l$. To obtain the first-order derivatives, we rewrite the optimization problem $\mathscr{P}$ in Lagrange form as

$$\mathscr{P}_l: \quad \underset{\boldsymbol{s}_l}{\text{minimize}} \quad \sum_{i=1}^{N_l+1} \left(\|\boldsymbol{x}_l^{(i)} - \boldsymbol{x}_l^{(i-1)}\|_2 + \gamma_l^{(i)}\, g_l^{(i)}(\boldsymbol{x}_l^{(i)}) \right) \tag{3.3}$$

where $\gamma_l^{(i)}$ denotes the Lagrange multiplier associated with the surface function $g_l^{(i)}$. Then, the KKT optimality conditions for the reflection point $\boldsymbol{x}_l^{(i)}$ corresponding to the path segment $(\boldsymbol{x}_l^{(i-1)}, \boldsymbol{x}_l^{(i)}, \boldsymbol{x}_l^{(i+1)})$ are given by

$$\begin{aligned} \mathbf{0} &= \frac{\partial \|\boldsymbol{x}_l^{(i)} - \boldsymbol{x}_l^{(i-1)}\|_2}{\partial \boldsymbol{x}_l^{(i)}} + \frac{\partial \|\boldsymbol{x}_l^{(i+1)} - \boldsymbol{x}_l^{(i)}\|_2}{\partial \boldsymbol{x}_l^{(i)}} + \gamma_l^{(i)} \frac{\partial g_l^{(i)}}{\partial \boldsymbol{x}_l^{(i)}} \\ 0 &= g_l^{(i)}(\boldsymbol{x}_l^{(i)})\,. \end{aligned} \tag{3.4}$$

For compactness, (3.4) can be written using Fermat's equation for the path segment as

$$\boldsymbol{g}_l^{(i)}\big(\boldsymbol{x}_l^{(i-1)}, \boldsymbol{x}_l^{(i+1)}, \boldsymbol{x}_l^{(i)}, \gamma_l^{(i)}\big) = \mathbf{0}\,. \tag{3.5}$$

Using the implicit function theorem, we obtain $\boldsymbol{\psi}_l^{(i)}$ as

$$\left[\, \boldsymbol{x}_l^{(i)} \;\; \gamma_l^{(i)} \,\right] = \boldsymbol{\psi}_l^{(i)}(\boldsymbol{x}_l^{(i-1)}, \boldsymbol{x}_l^{(i+1)}) \tag{3.6}$$

with Jacobian $\mathbf{J}_{\boldsymbol{\psi}_l^{(i)}}$ defined as

$$\mathbf{J}_{\boldsymbol{\psi}_l^{(i)}} = -\left[\frac{\partial \boldsymbol{g}_l^{(i)}}{\partial \boldsymbol{x}_l^{(i)}} \;\; \frac{\partial \boldsymbol{g}_l^{(i)}}{\partial \gamma_l^{(i)}} \right]^{-1} \left[\frac{\partial \boldsymbol{g}_l^{(i)}}{\partial \boldsymbol{x}_l^{(i-1)}} \;\; \frac{\partial \boldsymbol{g}_l^{(i)}}{\partial \boldsymbol{x}_l^{(i+1)}} \right]. \tag{3.7}$$

Note that at the ith bounce, (3.6) and (3.7) involve both the previous and the next reflection points. However, intuitively, since the BS is stationary, the gradient should propagate from the UE toward the BS through the reflection points. Next, we formalize this intuition by providing an induction proof. At bounce i, our goal is to express all quantities in terms of $\boldsymbol{x}_l^{(i+1)}$ so that, when the gradient is computed, it depends only on the reflection points $\boldsymbol{x}_l^{(i+1)}, \boldsymbol{x}_l^{(i+2)}, \cdots, \boldsymbol{x}_l^{(N_l)}$. Since the BS is fixed, the implicit function at the 1st bounce depends only on $\boldsymbol{x}_l^{(2)}$. Thus, it reduces to

$$\left[\, \boldsymbol{x}_l^{(1)} \;\; \gamma_l^{(1)} \,\right] = \boldsymbol{\psi}_l^{(1)}(\boldsymbol{x}_l^{(2)})\,. \tag{3.8}$$

Substituting (3.8) into Fermat's equation (3.5) at the 2nd bounce yields

$$\boldsymbol{g}_l^{(2)}\big(\boldsymbol{\psi}_l^{(1)}(\boldsymbol{x}_l^{(2)}), \boldsymbol{x}_l^{(3)}, \boldsymbol{x}_l^{(2)}, \gamma_l^{(2)}\big) = \mathbf{0}. \tag{3.9}$$

We assume that at bounce i, there exists a function $\boldsymbol{\psi}_l^{(i)}$ that takes $\boldsymbol{x}_l^{(i+1)}$ as input and maps it to $[\, \boldsymbol{x}_l^{(i)} \; \gamma_l^{(i)} \,]$. Then, at bounce $i+1$, we can write Fermat's equation as

$$\boldsymbol{g}_l^{(i+1)}\big(\boldsymbol{\psi}_{l,\boldsymbol{x}}^{(i)}(\boldsymbol{x}_l^{(i+1)}), \boldsymbol{x}_l^{(i+2)}, \boldsymbol{x}_l^{(i+1)}, \gamma_l^{(i+1)}\big) = \mathbf{0}\,. \tag{3.10}$$

By the implicit function theorem, there exists a function $\boldsymbol{\psi}_l^{(i+1)}$ that maps $\boldsymbol{x}_l^{(i+2)}$ to $[\, \boldsymbol{x}_l^{(i+1)} \; \gamma_l^{(i+1)} \,]$. This concludes our induction argument. Therefore,

$$\boldsymbol{g}_l^{(i)}\big(\boldsymbol{\psi}_{l,\boldsymbol{x}}^{(i-1)}(\boldsymbol{\psi}_{l,\boldsymbol{x}}^{(i)}(\boldsymbol{x}_l^{(i+1)})), \boldsymbol{x}_l^{(i+1)}, \boldsymbol{\psi}_l^{(i)}(\boldsymbol{x}_l^{(i+1)})\big) = \mathbf{0}\,. \tag{3.11}$$

By differentiating (3.11) with respect to $\boldsymbol{x}_l^{(i+1)}$, we obtain

$$0 = \frac{\partial \boldsymbol{g}_l^{(i)}}{\partial \boldsymbol{x}_l^{(i+1)}} + \frac{\partial \boldsymbol{g}_l^{(i)}}{\partial \boldsymbol{x}_l^{(i-1)}} \frac{\partial \boldsymbol{\psi}_{l,\boldsymbol{x}}^{(i-1)}}{\partial \boldsymbol{x}_l^{(i)}} \frac{\partial \boldsymbol{\psi}_{l,\boldsymbol{x}}^{(i)}}{\partial \boldsymbol{x}_l^{(i+1)}} \tag{3.12}$$

$$+ \frac{\partial \boldsymbol{g}_l^{(i)}}{\partial \boldsymbol{x}_l^{(i)}} \frac{\partial \boldsymbol{\psi}_{l,\boldsymbol{x}}^{(i)}}{\partial \boldsymbol{x}_l^{(i+1)}} + \frac{\partial \boldsymbol{g}_l^{(i)}}{\partial \gamma_l^{(i)}} \frac{\partial \boldsymbol{\psi}_{l,\gamma}^{(i)}}{\partial \boldsymbol{x}_l^{(i+1)}}\,. \tag{3.13}$$

Here $\boldsymbol{\psi}_{l,\gamma}^{(i)}$ denotes the component of the implicit function responsible for $\gamma_l^{(i)}$. Subsequently,

$$\left[\frac{\partial \boldsymbol{g}_l^{(i)}}{\partial \boldsymbol{x}_l^{(i)}} + \frac{\partial \boldsymbol{g}_l^{(i)}}{\partial \boldsymbol{x}_l^{(i-1)}} \frac{\partial \boldsymbol{\psi}_l^{(i-1)}}{\partial \boldsymbol{x}_l^{(i)}}, \frac{\partial \boldsymbol{g}_l^{(i)}}{\partial \gamma_l^{(i)}}\right] \frac{\partial \boldsymbol{\psi}_l^{(i)}}{\partial \boldsymbol{x}_l^{(i+1)}} = -\left[\frac{\partial \boldsymbol{g}_l^{(i)}}{\partial \boldsymbol{x}_l^{(i+1)}}\right]\,. \tag{3.14}$$

From here, computing $\hat{\boldsymbol{x}}_l^{(i)}$ is application of chain rule as

$$\frac{\partial \boldsymbol{\psi}_{l,\boldsymbol{x}}^{(i)}}{\partial \boldsymbol{x}^{\mathrm{UE}}} = \frac{\partial \boldsymbol{\psi}_{l,\boldsymbol{x}}^{(i)}}{\partial \boldsymbol{x}_l^{(i+1)}} \frac{\partial \boldsymbol{\psi}_{l,\boldsymbol{x}}^{(i+1)}}{\partial \boldsymbol{x}_l^{(i+2)}} \cdots \frac{\partial \boldsymbol{\psi}_{l,\boldsymbol{x}}^{(N_l)}}{\partial \boldsymbol{x}^{\mathrm{UE}}}\,. \tag{3.15}$$

3.2 Numerical Simulations

We consider a mmWave system where BS and UE are equipped with omnidirectional antennas. Two case studies are conducted in a two-dimensional scene measuring 30 meters by 30 meters. This setup is chosen to simplify our exposition and provide deeper insights into the methodology. The BS is located at $\boldsymbol{x}^{\mathrm{BS}} = [\,5\,\mathrm{m}\;\; 5\,\mathrm{m}\,]$, and the position of the UE at time t is $\boldsymbol{x}^{\mathrm{UE}} = [\,25\,\mathrm{m}\;\; 25\,\mathrm{m}\,]$. To simulate UE mobility, we consider a scenario in which the UE location varies randomly according to a Gaussian distribution with zero mean and standard deviation σ, i.e., $\boldsymbol{\Delta} \sim \mathcal{N}(\mathbf{0}_2, \sigma^2 \boldsymbol{I}_2)$. The relative permittivity of all the surfaces is $\eta = 1$. To assess the performance of the proposed DRT in terahertz (THz) and millimeter wave (mmWave) bands, we employ 30 GHz and 3 GHz bands. We compare the channel NMSE performance of the proposed DRT against conventional dynamic RT, where NMSE is defined as:

$$e_{\mathrm{NMS}} = \frac{\big\|\boldsymbol{H}_{\hat{\mathcal{C}}}(f) - \boldsymbol{H}_{\mathcal{C}}(f)\big\|_{\mathrm{F}}^2}{\big\|\boldsymbol{H}_{\mathcal{C}}(f)\big\|_{\mathrm{F}}^2} \tag{3.16}$$

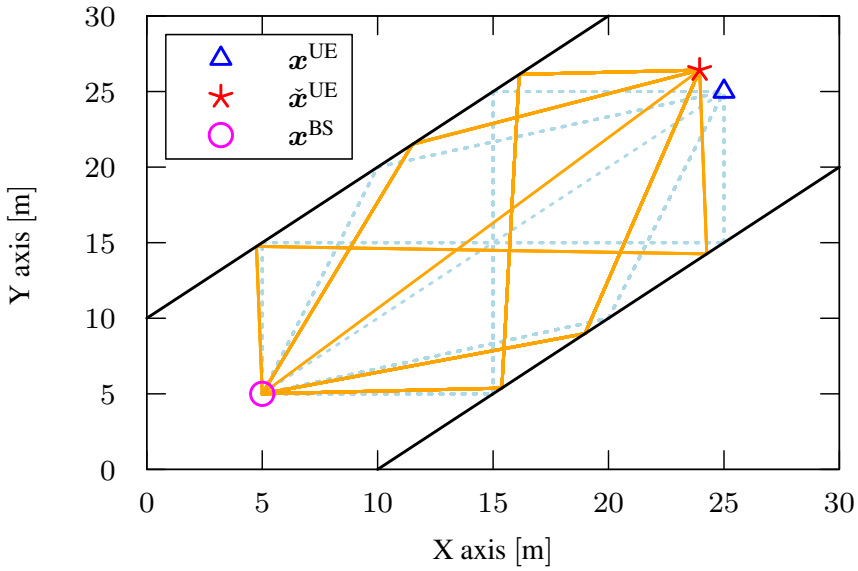


Figure 3.1: Setup for case study 1: planar surface.

3.2.1 Case Study 1: Planar Surface

In the first case study, we consider two parallel reflectors that resemble a simplified two-dimensional urban canyon model (see Fig. 3.1). There are no discontinuities along the individual reflectors. Fig.3.2 shows NMSE of the predicted CSI as a function of σ, and λ. It can be observed that, for the perturbation-based propagation path estimation method, the NMSE increases with the variance of $\boldsymbol{\Delta}$. This increase can be attributed to the growing error term in the Taylor-series approximation that is proportional to the displacement magnitude. The NMSE performance also depends on the wavelength. At shorter wavelengths, small errors in estimating reflection points lead to large changes in the phase of the estimated channel. Therefore, the NMSE at shorter wavelengths is higher than that at longer wavelengths. In comparison, the dynamic RT method exhibits stable NMSE performance across different wavelengths and displacement magnitudes. This robustness is primarily attributed to IM's capability to accurately estimate reflection points on planar surfaces.

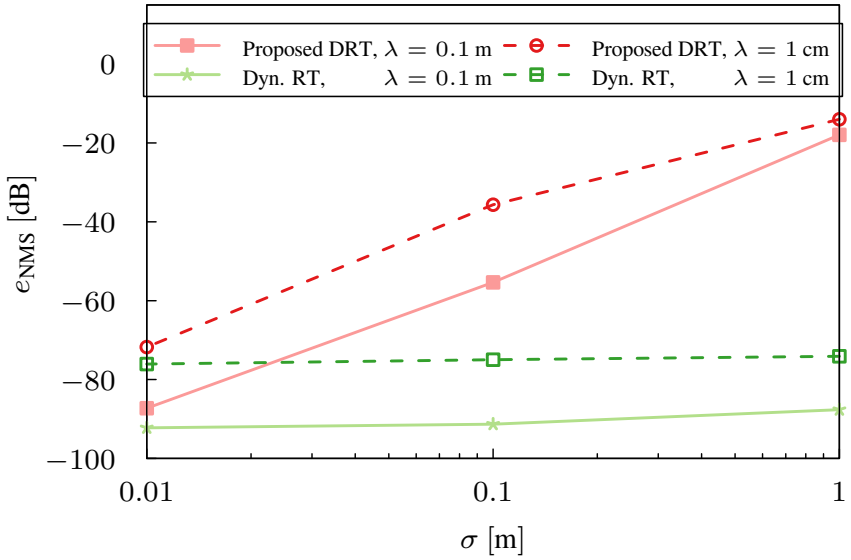


Figure 3.2: A plot of NMSE of estimated CSI for case study 1: planar surface.

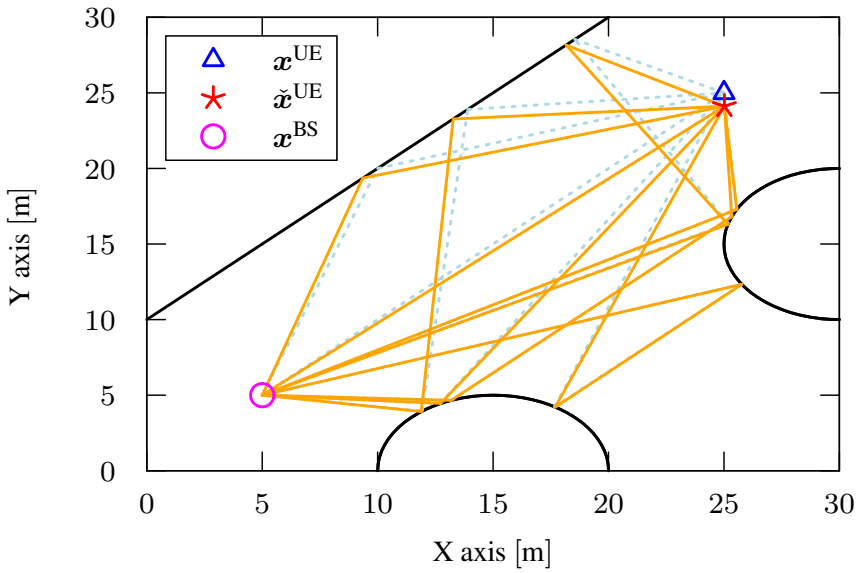


Figure 3.3: Setup for case study 2: curved surface.

3.2.2 Case Study 2: Curved Surface

The previous case study considered only planar surfaces with two parallel reflectors. In this case study, we examine two circular surfaces with centers located at [15 m 0 m] and [10 m 30 m], each having a radius of 5 m (see Fig. 3.3). For the dynamic RT method to

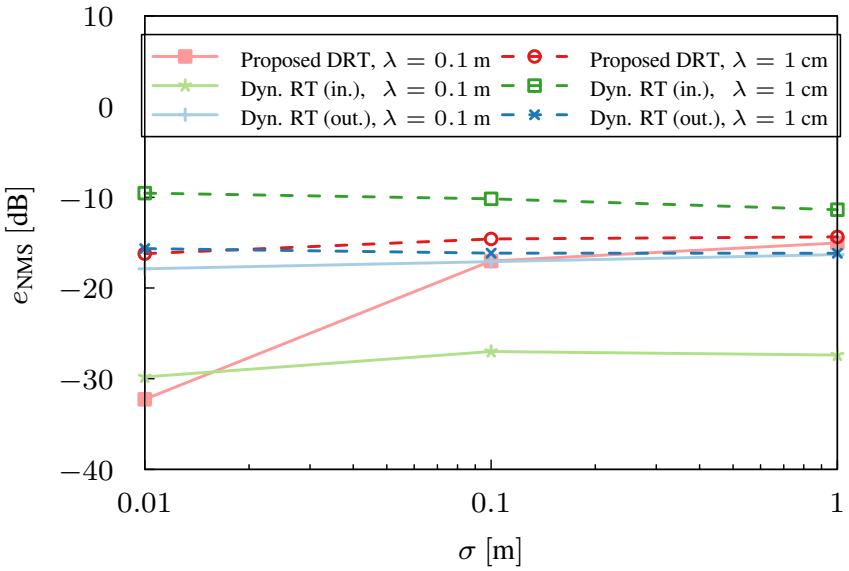


Figure 3.4: A plot of NMSE of estimated CSI for case study 2: curved surface.

operate, the circular surfaces are approximated using planar facets. Each planar facet spans an inscribed angle of 5°. Two different approximation strategies are simulated: (1) outer approximation, where planar faces are tangent to the circular surface (in Fig. 3.4 Dyn RT (in.)); and (2) inner approximation, where chords are drawn within the interior of the circular surface (in Fig. 3.4 Dyn RT (out.)).

Fig. 3.4 shows NMSE of the predicted CSI as a function of standard deviation σ, and wavelength λ. It can be observed that, in this scenario, the degradation of NMSE performance with decreasing wavelength is not particularly significant. However, a notable inversion in NMSE performance between the proposed method and dynamic RT can be observed. This improvement can be attributed to the greater generalizability of the proposed method. Since the proposed method does not rely on planar-surface assumptions to compute the Jacobian matrices, the predicted channel remains accurate, maintaining an NMSE below -30 dB. In contrast, the dynamic RT method exhibits inaccuracies arising from planar surface approximations, causing its estimated reflection points to deviate from the true reflection locations. Furthermore, in many cases, reflection points predicted by dynamic RT do not correspond to any actual reflection points on the curved surfaces.

3.2.3 Runtime Complexity

Tab. 3.1 presents the runtime of RT and the resulting NMSE in CSI estimation error incurred when approximating curved surfaces with planar segments. Here, Δ_θ denotes the inscribed angle of each plane, and the runtime is normalized against that observed without planar approximation. The CSI is computed at $f = 30$ GHz, and the NMSE is measured relative to the true CSI obtained without planar approximation. As the number of planes in the scene increases, the computational load on the RT solver grows accordingly. Thus, for smaller Δ_θ,

Table 3.1: NMSE and normalized runtime performance of inner and outer approximations.

	NMSE		**Normalized runtime**	
Δ_θ	**outer**	**inner**	**outer**	**inner**
5	0.0762	0.0223	2.22394	2.22394
10	0.0227	0.0226	1.37324	1.39429
20	0.0234	0.0229	0.955174	0.973745

the observed runtime is higher, approximately twice that of the non-approximated case. The runtime is normalized against the ray tracing time required to trace with circular shapes instead of planar approximations. Conversely, the normalized runtime decreases as the number of planes decreases. This NMSE serves as a lower bound on the performance of dynamic RT, since dynamic RT cannot eliminate the approximation error introduced by planarization of the reflector surface. Our observation that the inner approximation outperforms the outer approximation is consistent with earlier emulation results. The figure also suggests that using dynamic RT requires a careful trade-off between runtime and the NMSE of the predicted CSI. In contrast, the proposed DRT is not limited by such surface-approximation errors and achieves improved performance in both low-mobility and high-frequency regimes.

Chapter 4

Manifold Optimization-based Propagation Path Estimation

In this section, we describe the proposed DRT framework, built on the manifold optimization–based PPE (PPE-MO) technique. The key challenge in estimating propagation paths is the computational overhead associated with explicitly enforcing the geometric constraints of interacting surfaces. The proposed PPE-MO addresses this challenge by exploiting the smoothness of these surface constraints and incorporating them directly into the geometry of the search space by modelling each physical surface as a smooth submanifold. This formulation allows us to recast $\mathscr{P}_l$ as an unconstrained optimization problem over a manifold, where the surface constraints in (2.19b) are handled implicitly. Also, it eliminates the need for complex Lagrange multipliers and avoids the gradient drift associated with extrinsic penalty methods. Another key distinguishing feature of the proposed PPE-MO method is that, unlike MPT, it leverages the previous propagation paths $\boldsymbol{s}_l$ as a warm start, rather than recomputing the propagation paths from scratch. Consequently, the estimated paths $\hat{\boldsymbol{s}}_l$ can be obtained with fewer iterations by exploiting the smoothness of the constraint surfaces and the resulting proximity of the interaction points.

The remainder of this section describes the PPE-MO method for two different implementations of DTs: (1) continuous manifold-based DTs, and (2) discrete manifold-based DTs. Before delving into the details of PPE-MO, we first introduce the key concepts of manifold optimization and reformulate $\mathscr{P}_l$ within the framework of Riemannian geometry.

4.1 Preliminaries on Manifold Optimization

The surfaces that are typically present in a DT are smooth surfaces, and the optimization problem $\mathscr{P}_l$ is constrained over this set of smooth surfaces. In terms of manifold optimization, these smooth surface constraints in the optimization problem $\mathscr{P}$ define smooth submanifolds embedded in $\mathbb{R}^3$. Here, we provide a definition of smoothness of manifolds:

Definition 1. *A subset $\mathcal{M} \subset \mathbb{R}^3$ is a smooth embedded submanifold if, for every point $\boldsymbol{x} \in \mathcal{M}$, there exists a neighborhood $\mathcal{U} \subset \mathbb{R}^3$ of $\boldsymbol{x}$, an open set $\mathcal{V} \subset \mathbb{R}^3$, and a diffeomorphism $\psi : \mathcal{U} \to \mathcal{V}$ such that $\psi(\mathcal{U} \cap \mathcal{M}) = \mathcal{V} \cap (\mathbb{R}^2 \times \{0\})$.*

For a given chain l, the ith interaction point is constrained to the manifold $\mathcal{M}_l^{(i)}$, which is defined implicitly by the surface equation $g_l^{(i)}(\boldsymbol{x}_l^{(i)}) = 0$. Consequently, the complete sequence of interaction points comprising the lth chain, $\boldsymbol{s}_l$, resides within the product manifold $\mathcal{M}_l = \prod_{i=1}^{N_l} \mathcal{M}_l^{(i)}$. To characterize the local geometry at a specific interaction point $\boldsymbol{x}_l^{(i)}$, we look at the tangent vector at that point:

Definition 2. *Let $c : (-\epsilon, \epsilon) \to \mathcal{M}_l^{(i)}$ denote a smooth curve parametrized by $t \in \mathbb{R}$ with $c(0) = \boldsymbol{x}$. A tangent vector $c'(0)$ at $\boldsymbol{x}$ is defined as the velocity of this curve evaluated at $t = 0$.*

The collection of all such tangent vectors at $\boldsymbol{x}_l^{(i)}$ forms the tangent space, denoted by $T_{\boldsymbol{x}_l^{(i)}}\mathcal{M}_l^{(i)} = \left\{c'(0) \mid c : (-\epsilon, \epsilon) \to \mathcal{M}_l^{(i)} \text{ is smooth and } c(0) = \boldsymbol{x}_l^{(i)}\right\}$. Finally, the tangent bundle is defined as $T\mathcal{M}_l^{(i)} = \left\{(\boldsymbol{x}, \boldsymbol{v}) : \boldsymbol{x} \in \mathcal{M}_l^{(i)}, \boldsymbol{v} \in T_{\boldsymbol{x}}\mathcal{M}_l^{(i)}\right\}$.

Since the reflecting surfaces of the scatterers are smoothly embedded in $\mathbb{R}^3$, they naturally inherit a Riemannian geometry from the ambient space. For any two tangent vectors $\boldsymbol{u}, \boldsymbol{v} \in T_{\boldsymbol{x}_l^{(i)}}\mathcal{M}_l^{(i)}$, the Riemannian metric is defined by the induced Euclidean inner product: $\langle \boldsymbol{u}, \boldsymbol{v} \rangle_{\boldsymbol{x}_l^{(i)}} = \boldsymbol{u}^{\mathrm{T}}\boldsymbol{v}$. Under this Riemannian metric, the distance function in the objective of $\mathscr{P}_l$ varies smoothly with respect to the interaction points. Thus, a natural approach to solve $\mathscr{P}_l$ is to employ a gradient-based method. To determine the descent direction that is compatible with the manifold geometry, we use the notion of the Riemannian gradient, defined as

Definition 3. *The Riemannian gradient* $\operatorname{grad} f$ *of a smooth function $f : \mathcal{M}_l^{(i)} \to \mathbb{R}$ is the vector field defined by*

$$(\operatorname{grad} f(\boldsymbol{x}))^{\mathrm{T}}\boldsymbol{v} = Df(\boldsymbol{x})[\boldsymbol{v}], \quad \forall (\boldsymbol{x}, \boldsymbol{v}) \in T\mathcal{M}_l^{(i)}. \tag{4.1}$$

Here, $Df(\boldsymbol{x})[\boldsymbol{v}]$ denotes the differential of f at $\boldsymbol{x}$. In other words, the Riemannian gradient computed on the manifold $\mathcal{M}_l^{(i)}$ is the unique tangent vector at $T_{\boldsymbol{x}}$ that points to the steepest descent direction of f at $\boldsymbol{x}$. We utilize the standard Euclidean distance $\|\cdot\|_2$ of the ambient space $\mathbb{R}^3$ to define spatial proximity. This ambient metric allows us to define a projection operator onto the manifolds.

For any point $\boldsymbol{x} \in \mathbb{R}^3$, the projection $\operatorname{proj}_{\mathcal{M}_l^{(i)}}(\boldsymbol{x})$ of $\boldsymbol{x}$ on a manifold $\mathcal{M}_l^{(i)}$ is the closest point on $\mathcal{M}_l^{(i)}$ from $\boldsymbol{x}$, i.e.,

$$\operatorname{proj}_{\mathcal{M}_l^{(i)}}(\boldsymbol{x}) = \left\{\boldsymbol{y} \in \mathcal{M}_l^{(i)} : \|\boldsymbol{x} - \boldsymbol{y}\|_2 = \inf_{\boldsymbol{z} \in \mathcal{M}_l^{(i)}} \|\boldsymbol{x} - \boldsymbol{z}\|_2\right\}. \tag{4.2}$$

The projection operator is typically used when an update step causes a point to leave the manifold, projecting it back onto the surface by minimizing the distance to the manifold. However, computing this projection often requires solving an additional optimization problem to determine the point that achieves the infimum distance. To avoid this computational overhead, retraction is introduced as a more efficient alternative. The operation of retraction is defined as follows:

Definition 4. *A retraction* $\mathrm{retr}_{\boldsymbol{x}}$ *at* $\boldsymbol{x} \in \mathcal{M}_l^{(i)}$ *is a smooth map* $\mathrm{retr}_{\boldsymbol{x}} : T_{\boldsymbol{x}}\mathcal{M} \to \mathcal{M}$ *such that* $(\boldsymbol{x}, \boldsymbol{v}) \to R_{\boldsymbol{x}}(\boldsymbol{v})$ *where each curve* $c(t) = \mathrm{retr}_{\boldsymbol{x}}(t\boldsymbol{v})$ *satisfies* $c(0) = \boldsymbol{x}$ *and* $c'(0) = \boldsymbol{v}$.

Retraction provides a smooth mapping that moves a point along a tangent direction back onto the manifold without requiring an explicit projection step. Retraction can be used to compute the Riemannian gradient at a point $\boldsymbol{x}$ as follows:

$$(\mathrm{grad}\, f(\boldsymbol{x}))^{\mathrm{T}}\boldsymbol{v} = \frac{\mathrm{d}}{\mathrm{d}t}\big(f \circ \mathrm{retr}_{\boldsymbol{x}}(t\boldsymbol{v})\big) \mid_{t=0} . \tag{4.3}$$

Another common operation in manifold optimization involves mapping vectors between different vector spaces. In particular, metric projection can be used to map an arbitrary vector onto a tangent space. This mapping from an arbitrary vector to the tangent space of a manifold is performed using the orthogonal projection operator. The orthogonal projection of a vector $\boldsymbol{v} \in \mathbb{R}^3$ onto the tangent space $T_{\boldsymbol{x}}\mathcal{M}$ at $\boldsymbol{x} \in \mathcal{M}$ is the unique vector $\mathbf{P}_{T_{\boldsymbol{x}}\mathcal{M}}(\boldsymbol{v}) \in T_{\boldsymbol{x}}\mathcal{M}$ satisfying

$$(\boldsymbol{v} - \mathbf{P}_{T_{\boldsymbol{x}}\mathcal{M}}(\boldsymbol{v}))^{\mathrm{T}}\boldsymbol{d} = 0 \tag{4.4}$$

for any $\boldsymbol{d} \in T_{\boldsymbol{x}}\mathcal{M}$.

To illustrate these definitions, we consider the example of the surface of a unit sphere represented by the manifold:

$$\mathcal{M} = \{\, \boldsymbol{x} : \|\boldsymbol{x}\|_2 = 1 \,\}. \tag{4.5}$$

At any point $\boldsymbol{x} \in \mathcal{M}$, the tangent space is:

$$T_{\boldsymbol{x}}\mathcal{M} = \{\, \boldsymbol{v} \in \mathbb{R}^3 : \boldsymbol{v}^{\mathrm{T}}\boldsymbol{x} = 0\}. \tag{4.6}$$

Note that the retraction operation is not unique. One example of a retraction operator for a tangent vector $\boldsymbol{v} \in T_{\boldsymbol{x}}\mathcal{M}$ is:

$$\mathrm{retr}_{\boldsymbol{x}}(\boldsymbol{v}) = \frac{\boldsymbol{x} + \boldsymbol{v}}{\|\boldsymbol{x} + \boldsymbol{v}\|_2}. \tag{4.7}$$

The orthogonal projection of a vector $\boldsymbol{v} \in \mathbb{R}^3$ is

$$\mathbf{P}_{T_{\boldsymbol{x}}\mathcal{M}}(\boldsymbol{v}) = \boldsymbol{v} - (\boldsymbol{v}^{\mathrm{T}}\boldsymbol{x})\,\boldsymbol{x}. \tag{4.8}$$

Next, we consider a smooth function f:

$$f(\boldsymbol{x}) = \|\boldsymbol{x} - \boldsymbol{p}\|_2 + \|\boldsymbol{q} - \boldsymbol{x}\|_2. \tag{4.9}$$

This function is particularly interesting as it quantifies the length of a one-bounce path. From (4.3), we have

$$\begin{aligned}
&(\mathrm{grad}\, f(\mathbf{x}))^{\mathrm{T}}\boldsymbol{v} \\
&\quad= \frac{\mathrm{d}}{\mathrm{d}t}\left[\left\|\frac{\boldsymbol{x} + t\boldsymbol{v}}{\|\boldsymbol{x} + t\boldsymbol{v}\|_2} - \boldsymbol{p}\right\|_2 + \left\|\boldsymbol{q} - \frac{\boldsymbol{x} + t\boldsymbol{v}}{\|\boldsymbol{x} + t\boldsymbol{v}\|_2}\right\|_2\right]_{t=0} \\
&\quad= \left(\frac{(\boldsymbol{x} - \boldsymbol{p})}{\|\boldsymbol{x} - \boldsymbol{p}\|_2} + \frac{(\boldsymbol{x} - \boldsymbol{q})}{\|\boldsymbol{x} - \boldsymbol{q}\|_2}\right)^{\mathrm{T}} \big(\boldsymbol{v} - (\boldsymbol{v}^{\mathrm{T}}\boldsymbol{x})\,\boldsymbol{x}\big)
\end{aligned} \tag{4.10}$$

Therefore, the Riemannian gradient of f is given by

$$\mathrm{grad}\, f(\boldsymbol{x}) = (\boldsymbol{I} - \boldsymbol{x}\boldsymbol{x}^{\mathrm{T}})\left(\frac{(\boldsymbol{x} - \boldsymbol{p})}{\|\boldsymbol{x} - \boldsymbol{p}\|_2} + \frac{(\boldsymbol{x} - \boldsymbol{q})}{\|\boldsymbol{x} - \boldsymbol{q}\|_2}\right). \tag{4.11}$$

4.2 PPE Problem Reformulation

Within the framework of manifold optimization, $\mathscr{P}_l$ is reformulated as

$$\tilde{\mathscr{P}}_l: \quad \underset{\boldsymbol{s}_l \in \mathcal{M}_l}{\text{minimize}} \quad f_l(\boldsymbol{s}_l, \boldsymbol{x}^{\text{UE}}) \tag{4.12}$$

where $f_l(\boldsymbol{s}_l, \boldsymbol{x}^{\text{UE}})$ is the objective function defined as

$$f_l(\boldsymbol{s}_l, \boldsymbol{x}^{\text{UE}}) = \sum_{i=1}^{N_l+1} \left\| \boldsymbol{x}_l^{(i)} - \boldsymbol{x}_l^{(i-1)} \right\|_2 . \tag{4.13}$$

Note that f_l admits a smooth extension to the ambient space $\mathbb{R}^{3\times(N_l+1)}$. Let the optimal solution of (4.12) be denoted by $\boldsymbol{s}_l^\star$. Also, let $\boldsymbol{u}_l : \mathcal{M}_l \times \mathcal{U} \to T\mathcal{M}_l$ denote the gradient vector field of the objective function, defined as a section of the tangent bundle over the product space of $\mathcal{M}_l$ and the UE position. Then, the first-order optimality condition requires that the gradient function vanishes at the optimal solution $\boldsymbol{s}_l^\star$:

$$\boldsymbol{u}_l(\boldsymbol{s}_l^\star, \boldsymbol{x}^{\text{UE}}) \triangleq \text{grad}\, f_l(\boldsymbol{s}_l^\star, \boldsymbol{x}^{\text{UE}}) = \boldsymbol{0} . \tag{4.14}$$

Geometrically, (4.14) implies that the optimal solution $\boldsymbol{s}_l^\star$ lies at the intersection of the section $\boldsymbol{u}_l$ and the zero section of the tangent bundle. As the UE moves from $\boldsymbol{x}^{\text{UE}}$ to $\check{\boldsymbol{x}}^{\text{UE}}$, this optimality condition must be satisfied continuously as the interaction points change from $\boldsymbol{s}_l$ to $\check{\boldsymbol{s}}_l$. For small perturbations in the UE position $\Delta\boldsymbol{x}^{\text{UE}} = \check{\boldsymbol{x}}^{\text{UE}} - \boldsymbol{x}^{\text{UE}}$, the corresponding shift in the interaction path can be derived by linearizing the gradient vector field around the stationary point $\boldsymbol{s}_l$ as

$$\begin{aligned} \boldsymbol{u}_l(\check{\boldsymbol{s}}_l, \check{\boldsymbol{x}}^{\text{UE}}) \approx\ & \boldsymbol{u}_l(\boldsymbol{s}_l, \boldsymbol{x}^{\text{UE}}) \\ & + \nabla_{\boldsymbol{s}_l} \boldsymbol{u}_l \left(\check{\boldsymbol{s}}_l - \boldsymbol{s}_l\right) + \nabla_{\boldsymbol{x}^{\text{UE}}} \boldsymbol{u}_l \Delta\boldsymbol{x}^{\text{UE}} . \end{aligned} \tag{4.15}$$

Because the unperturbed state is already optimal, we have $\boldsymbol{u}_l(\boldsymbol{s}_l, \boldsymbol{x}^{\text{UE}}) = 0$. By enforcing that the perturbed state must also remain optimal, i.e., the left side of the expansion in (4.15) must also equal zero, we obtain the differential relation:

$$\nabla_{\boldsymbol{s}_l} \boldsymbol{u}_l \left(\check{\boldsymbol{s}}_l - \boldsymbol{s}_l\right) + \nabla_{\boldsymbol{x}^{\text{UE}}} \boldsymbol{u}_l \, \Delta\boldsymbol{x}^{\text{UE}} = \boldsymbol{0} \tag{4.16}$$

We note that (4.16) requires the Riemannian Hessian of f_l to be non-singular to solve for the path update $\Delta\boldsymbol{s}_l$. This Hessian is naturally non-singular provided the rays are not caustic, and the interaction points are within the interior of the surfaces. Therefore, under the small UE movement assumption, the problem $\tilde{\mathscr{P}}_l$ can be reformulated as solving the set of linear equations in (4.16) to find $\hat{\boldsymbol{s}}_l$ as

$$\check{\mathscr{P}}_l: \quad \underset{\hat{\boldsymbol{s}}_l \in \mathcal{M}_l}{\text{minimize}} \quad \left\| \nabla_{\boldsymbol{s}_l} \boldsymbol{u}_l \left(\hat{\boldsymbol{s}}_l - \boldsymbol{s}_l\right) + \nabla_{\boldsymbol{x}^{\text{UE}}} \boldsymbol{u}_l \, \Delta\boldsymbol{x}^{\text{UE}} \right\|_2 . \tag{4.17}$$

Note that the accuracy of the solution of $\check{\mathscr{P}}_l$ relies on the numerical precision of the computed Hessian $\nabla_{\boldsymbol{x}^{\text{UE}}} \boldsymbol{u}_l$, as any error here accumulates in the predicted interaction chain. Such numerical inaccuracies arise from floating-point limitations or ill-conditioning when evaluating second-order derivatives, especially for multi-bounce paths where minor geometric perturbations are heavily amplified in the interaction chain. To address this issue, we first establish a fundamental geometric property of $\nabla_{\boldsymbol{x}^{\text{UE}}} \boldsymbol{u}_l$ in the following proposition, which allows us to isolate the effective components of the UE displacement.

Proposition 1. *For the interaction points $\boldsymbol{s}_l$ and UE position $\boldsymbol{x}^{\mathrm{UE}}$, the gradient vector field at $\boldsymbol{x}^{\mathrm{UE}}$ is orthogonal to the direction vector $\boldsymbol{k}_l^{\mathrm{UE}}$:*

$$\boldsymbol{k}_l^{\mathrm{UE}} \in \mathrm{null}(\nabla_{\boldsymbol{x}^{\mathrm{UE}}} \boldsymbol{u}_l). \tag{4.18}$$

Proof. For any point $\boldsymbol{x}$ of the form $\boldsymbol{x} = \boldsymbol{x}^{\mathrm{UE}} + t\,\boldsymbol{k}_l^{\mathrm{UE}}$ for some t in a small centered neighborhood $t \in (-\epsilon, \epsilon)$, f_l can be written as

$$\begin{aligned} f_l(\boldsymbol{s}_l, \boldsymbol{x}) &= \sum_{i=1}^{N_l} \|\boldsymbol{x}_l^{(i)} - \boldsymbol{x}_l^{(i-1)}\|_2 + \|\boldsymbol{x}_l^{(N_l)} - \boldsymbol{x}\|_2 \\ &= \sum_{i=1}^{N_l} \|\boldsymbol{x}_l^{(i)} - \boldsymbol{x}_l^{(i-1)}\|_2 + \|\boldsymbol{x}_l^{(N_l)} - \boldsymbol{x}^{\mathrm{UE}} - t\,\boldsymbol{k}_l^{\mathrm{UE}}\|_2 \\ &= f_l(\boldsymbol{s}_l, \boldsymbol{x}^{\mathrm{UE}}) - t\,. \end{aligned} \tag{4.19}$$

Note that (4.19) holds as $\boldsymbol{x}$ and $\boldsymbol{x}^{\mathrm{UE}}$ are on the same line: $\boldsymbol{x} = \boldsymbol{x}^{\mathrm{UE}} + t\,\boldsymbol{k}_l^{\mathrm{UE}}$. Since t does not depend on $\boldsymbol{s}_l$, $\mathrm{grad}\, f_l(\boldsymbol{s}_l, \boldsymbol{x}^{\mathrm{UE}}) = 0$ implies that $\mathrm{grad}\, f_l(\boldsymbol{s}_l, \boldsymbol{x}) = 0$. Because this holds for any value of t, $\mathrm{grad}\, f_l$ is constant in the direction $\boldsymbol{k}_l^{\mathrm{UE}}$. □

Consequently, displacements of the UE along $\boldsymbol{k}_l^{\mathrm{UE}}$ do not modify $\boldsymbol{s}_l$, and only the components of $\Delta\boldsymbol{x}^{\mathrm{UE}}$ that are orthogonal to $\boldsymbol{k}_l^{\mathrm{UE}}$ contribute to changes in the interaction points. We formalize this observation in the following remark.

Remark 1. *The effective UE displacement that is robust to the Hessian computation error is*

$$\left[\Delta\boldsymbol{x}^{\mathrm{UE}}\right]_{\perp} = \check{\boldsymbol{x}}^{\mathrm{UE}} - \boldsymbol{x}^{\mathrm{UE}} - \langle \check{\boldsymbol{x}}^{\mathrm{UE}} - \boldsymbol{x}^{\mathrm{UE}}, \boldsymbol{k}_l^{\mathrm{UE}} \rangle\, \boldsymbol{k}_l^{\mathrm{UE}}\,. \tag{4.20}$$

Following the remark, we reformulate the optimization problem as

$$\check{\mathscr{P}}_l : \underset{\hat{\boldsymbol{s}}_l \in \mathcal{M}_l}{\mathrm{minimize}} \left\| \nabla_{\boldsymbol{s}_l} \boldsymbol{u}_l\, (\hat{\boldsymbol{s}}_l - \boldsymbol{s}_l) + \nabla_{\boldsymbol{x}^{\mathrm{UE}}} \boldsymbol{u}_l \left[\Delta\boldsymbol{x}^{\mathrm{UE}}\right]_{\perp} \right\|_2 . \tag{4.21}$$

4.2.1 Continuous Manifold-based Digital Twin

Continuous manifold-based DTs are typically composed of, or constructed from, smooth geometric primitives. A geometric primitive (e.g., planes, spheres, cones) is a fundamental object that serves as a building block in DTs. Frameworks such as Mitsuba [21] and NVIDIA Omniverse support both curved and planar surfaces, enabling the representation of complex geometries. More intricate structures are formed by combining these primitives using constructive solid geometry. In this setting, each interacting surface in an interaction chain corresponds to a geometric primitive, with its shape defined by an associated surface equation. Since these primitives are smooth, gradient-based optimization methods can be effectively employed to obtain $\hat{\boldsymbol{s}}_l$ from (4.21). In order to use gradient-based optimization methods, we consider a variational formulation of (4.21) as follows:

$$\mathscr{P}_{\mathrm{CM},l} : \quad \underset{\hat{\boldsymbol{s}}_l \in \mathcal{M}_l}{\mathrm{minimize}} \quad f_{\mathrm{CM},l}(\hat{\boldsymbol{s}}_l) \tag{4.22}$$

Algorithm 1 Manifold Optimization-based PPE in Continuous Manifold-based DT

1: Choose initial point $\boldsymbol{s}_{l,0} \in \mathcal{M}_l$ as described and set $\eta_0 := -\text{grad}\, f_{\text{CM},l}(\boldsymbol{s}_{l,0})$ and $k := 0$.
2: **while** $\text{grad}\, f_{\text{CM},l}(\boldsymbol{s}_{l,k}) \neq 0$ **do**
3: Compute a step length $t_k > 0$ and $\boldsymbol{s}_{l,k+1} := \text{retr}_{\boldsymbol{s}_{l,k}}(t_k \boldsymbol{\eta}_k)$.
4: Compute $\beta_{k+1} = \frac{\|\text{grad}\, f_{\text{CM},l}(\boldsymbol{s}_{l,k+1})\|_2^2}{\|\text{grad}\, f_{\text{CM},l}(\boldsymbol{s}_{l,k})\|_2^2}$.
5: Compute a search direction as

$$\boldsymbol{\eta}_{k+1} \;=\; -\text{grad}\, f_{\text{CM},l}(\boldsymbol{s}_{l,k+1}) + \beta_{k+1} \mathbf{P}_{T_{\boldsymbol{s}_{l,k+1}} \mathcal{M}_l} \boldsymbol{\eta}_k$$

6: $k := k + 1$.
7: **end while**

where the objective function $f_{\text{CM},l}$ in $\mathscr{P}_{\text{CM},l}$ is:

$$f_{\text{CM},l}(\hat{\boldsymbol{s}}_l) = \|\nabla_{\boldsymbol{s}_l} \boldsymbol{u}_l \, (\hat{\boldsymbol{s}}_l - \boldsymbol{s}_l) + \nabla_{\boldsymbol{x}^{\text{UE}}} \boldsymbol{u}_l \, \left[\Delta \boldsymbol{x}^{\text{UE}}\right]_\perp \|_2 \,. \tag{4.23}$$

We note that the Riemannian gradients and Hessians are computed at the stationary point $\boldsymbol{s}_l$ obtained from the RT solution. The updates in interaction points are computed such that they remain on the product manifold $\mathcal{M}_l$. This thesis considers RCG to solve the optimization problem $\mathscr{P}_{\text{CM},l}$. RCG is an iterative method that takes conjugate gradient descent steps on the product manifold $\mathcal{M}_l$. At its core, RCG is governed by two update equations:

$$\boldsymbol{s}_{l,k+1} \;=\; \text{retr}_{\boldsymbol{s}_{l,k}}(t_k \boldsymbol{\eta}_k) \tag{4.24}$$

$$\boldsymbol{\eta}_{k+1} \;=\; -\text{grad}\, f_{\text{CM},l}(\boldsymbol{s}_{l,k+1}) + \beta_{k+1} \mathbf{P}_{T_{\boldsymbol{s}_{l,k+1}} \mathcal{M}_l}(\boldsymbol{\eta}_k) \,. \tag{4.25}$$

In the following, we describe the key operations involved in the RCG update equations. The gradient descent step is (4.24), where t_k and $\boldsymbol{\eta}_k$ are the step size and the gradient descent directions from step k, respectively. We note that taking a gradient-descent step might result in leaving the product manifold $\mathcal{M}_l$, as the surfaces are curved and the gradient step only approximates a linear change locally. To circumvent this issue, the updated interaction points are pulled back to the original surfaces using the retraction operation at $\boldsymbol{s}_{l,k}$. Typically, standard retraction operators for common smooth geometric primitives can be preprogrammed into the DT. Additionally, if a retraction operator is not available for a particular primitive, the metric projection onto the manifold from the ambient space can be used to realize the retraction operation. The choice of step size t_k depends on the Riemannian version of the Wolfe condition [22], and is obtained by a line search. The gradient descent direction is initialized with $\boldsymbol{\eta}_1 = -\text{grad}\, f_{\text{CM},l}(\boldsymbol{s}_{l,1})$.

In RCG, the main challenge is that the tangent space changes as the interaction points iterate over the manifold. Therefore, to compute the updated search direction in (4.25), the search direction $\boldsymbol{\eta}_k$ at step k needs to be transported to the tangent space of $\boldsymbol{s}_{l,k+1}$. This transformation is done using the orthogonal projection $\mathbf{P}_{T_{\boldsymbol{s}_{l,k}} \mathcal{M}_l}$. Lastly, the Fletcher-Reeves parameter β_{k+1} scales the transported search direction with:

$$\beta_{k+1} = \frac{\|\text{grad}\, f_{\text{CM},l}(\boldsymbol{s}_{l,k+1})\|_2^2}{\|\text{grad}\, f_{\text{CM},l}(\boldsymbol{s}_{l,k})\|_2^2} \,. \tag{4.26}$$

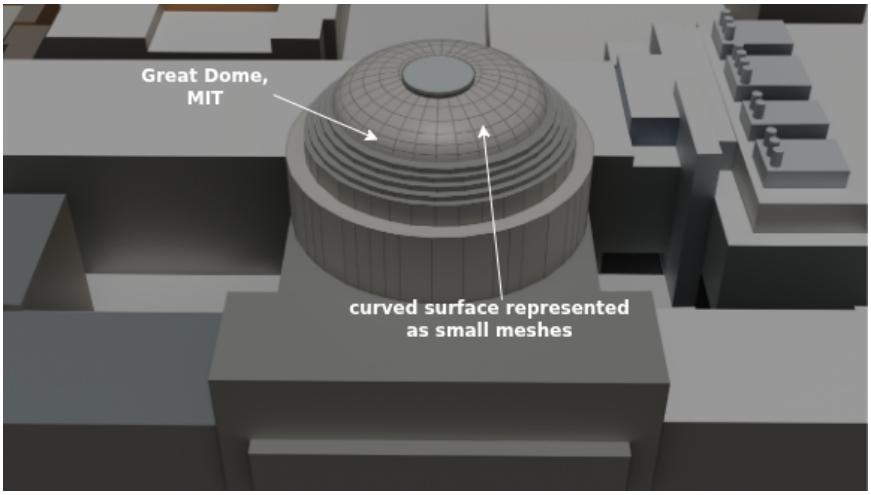


Figure 4.1: Discrete mesh representation of Great Dome at MIT.

Key steps of the algorithm are summarized in the listing Algorithm 1. The optimization algorithm requires an initial feasible point to begin the iterative process. Note that the interaction points $\boldsymbol{s}_l$ already satisfy the constraints in (2.19b), although they may be suboptimal for the updated UE position $\check{\boldsymbol{x}}^{\mathrm{UE}}$. Therefore, we use $\boldsymbol{s}_l$ for initialization of the algorithm.

4.2.2 Discrete Manifold-based Digital Twin

While continuous manifold-based DT provides well-defined gradients and retraction operators that work in conjunction with RCG to estimate propagation paths, several DTs do not directly support continuous manifolds (see Fig. 4.1). In such cases, the idealized smooth manifolds that characterize physical structures are instead realized as discrete triangular meshes to improve computational efficiency. In a discrete manifold-based DT, a continuous physical surface is approximated by a finite set of interconnected triangles, defined strictly by their vertices and normal vectors. Due to this discrete representation, gradients are no longer continuous at submanifold boundaries, causing the RCG method to cease to function effectively. Therefore, to perform continuous manifold optimization across the triangular mesh, the standard smooth retraction operator must be adapted to algorithmically handle boundary crossings. In the following, we first describe the mesh structure and the associated retraction operation. We then present the PPE-MO procedure for discrete manifold-based DT.

For a continuous manifold $\mathcal{M}$, the discrete triangular mesh is structured as a simplicial 2-complex, represented by the tuple $(\mathcal{V}, \mathcal{E}, \mathcal{F})$. Here, $\mathcal{V}$ denotes the set of spatial vertices, $\mathcal{E}$ represents the set of edges, and $\mathcal{F}$ represents the set of interconnected triangular faces. An arbitrary face $\iota \in \mathcal{F}$ is characterized by a triplet of vertices, corresponding to the spatial coordinates $(\boldsymbol{v}_{\iota,1}, \boldsymbol{v}_{\iota,2}, \boldsymbol{v}_{\iota,3})$. The topological connectivity of these simplices is encoded via two adjacency matrices: one mapping edges to their constituent vertices, and another mapping faces to their bounding edges. To parameterize the local geometry of a face ι, any point $\boldsymbol{x}$ residing on the plane defined by the face can be uniquely expressed via its barycentric

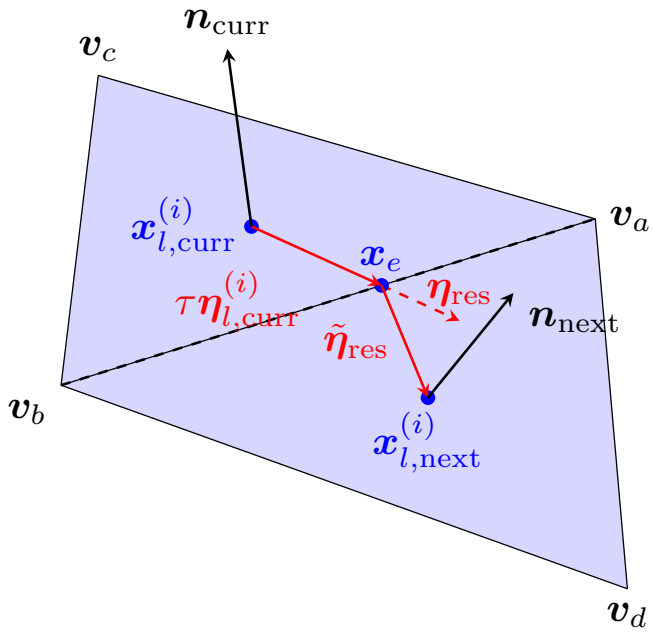


Figure 4.2: A diagram depicting a discrete mesh retraction process. The residual gradient is retracted back to the neighboring manifold using projection on the triangular mesh.

coordinates $[u, v, w]$ such that:

$$\boldsymbol{x} = u\,\boldsymbol{v}_{\iota,1} + v\,\boldsymbol{v}_{\iota,2} + w\,\boldsymbol{v}_{\iota,3}\,, \tag{4.27}$$

subject to the strict partition of unity constraint $u+v+w = 1$. The spatial point $\boldsymbol{x}$ lies strictly within the interior or on the edges of the triangular face ι if and only if the non-negativity conditions $u \geq 0$, $v \geq 0$, and $w \geq 0$ are simultaneously satisfied. The barycentric coordinates provide a computationally efficient, intrinsic mechanism to map continuous gradient steps onto the discrete piecewise-linear domain, allowing for the explicit algebraic detection of boundary crossings when any single barycentric weight becomes negative.

Next, we consider the discrete retraction process. We consider the interaction point $\boldsymbol{x}^{(i)}_l$ on the continuous manifold $\mathcal{M}^{(i)}_l$. Let $\boldsymbol{\eta}^{(i)}_{l,\text{curr}} \in T_{\boldsymbol{x}^{(i)}_l}\iota_{\text{curr}}$ denote a gradient descent step along a local tangent vector on the current face ι_{curr}. Note that ι_{curr} is part of the discretized mesh of the continuous manifold $\mathcal{M}^{(i)}_l$. If the gradient step computed on the current triangle does not cross the boundaries of the face, then the updated interaction point is

$$\boldsymbol{x}^{(i)}_{l,\text{next}} = \boldsymbol{x}^{(i)}_{l,\text{curr}} + \boldsymbol{\eta}^{(i)}_{l,\text{curr}}\,. \tag{4.28}$$

Otherwise, a discrete geometric unfolding technique is employed when the gradient step exceeds the spatial bounds of the current triangular face ι_{curr}. This unfolding technique is pictorially depicted in Fig. 4.2. Since the gradient descent step exceeds the strict barycentric bounds of ι_{curr}, the trajectory intersects a shared edge $e = \iota_{\text{curr}} \cap \iota_{\text{next}}$ at a boundary point $\boldsymbol{x}_{\text{e}} = \boldsymbol{x}^{(i)}_{l,\text{curr}} + \tau\boldsymbol{\eta}^{(i)}_{l,\text{curr}}$, where $\tau \in (0, 1)$ is the scalar fraction of the traversed path. The residual tangent vector to be mapped onto the adjacent face ι_{next} is

$$\boldsymbol{\eta}_{\text{res}} = (1-\tau)\boldsymbol{\eta}^{(i)}_{l,\text{curr}}\,. \tag{4.29}$$

To trace this remaining trajectory, ι_{next} is isometrically unfolded to be coplanar with ι_{curr}. Let $\boldsymbol{n}_{\text{curr}}$ and $\boldsymbol{n}_{\text{next}}$ denote the unit normals of ι_{curr} and ι_{next}, with the dihedral angle:

$$\theta = \arccos(\langle \boldsymbol{n}_{\text{curr}}, \boldsymbol{n}_{\text{next}} \rangle) \,. \tag{4.30}$$

The shared edge e, defined by vertices $\boldsymbol{v}_a$ and $\boldsymbol{v}_b$, serves as the axis of rotation with unit directional vector:

$$\boldsymbol{k} = \frac{\boldsymbol{v}_b - \boldsymbol{v}_a}{\|\boldsymbol{v}_b - \boldsymbol{v}_a\|_2} \,. \tag{4.31}$$

The isometric mapping of $\boldsymbol{\eta}_{\text{res}}$ onto the tangent space of the adjacent face ι_{next} is obtained using the Rodrigues' rotation R [23]:

$$\begin{aligned} \tilde{\boldsymbol{\eta}}_{\text{res}} &= R(\boldsymbol{\eta}_{\text{res}}; \theta, \boldsymbol{k}) \\ &= \boldsymbol{\eta}_{\text{res}} \cos\theta + (\boldsymbol{k} \times \boldsymbol{\eta}_{\text{res}}) \sin\theta + \boldsymbol{k}(\langle \boldsymbol{k}, \boldsymbol{\eta}_{\text{res}} \rangle)(1 - \cos\theta). \end{aligned} \tag{4.32}$$

The retracted coordinate on the adjacent face ι_{next} is

$$\boldsymbol{x}^{(i)}_{l,\text{next}} = \boldsymbol{x}_{\text{e}} + \tilde{\boldsymbol{\eta}}_{\text{res}}. \tag{4.33}$$

If $\boldsymbol{x}^{(i)}_{l,\text{next}}$ crosses the boundaries of the face ι_{next}, the unfolding technique is repeated. In general, at the kth step of unfolding, with $\boldsymbol{x}_{\text{e},k}$ denoting the point at edge with $\boldsymbol{x}_{\text{e},k} = \boldsymbol{x}^{(i)}_{l,k} + \tau_k \boldsymbol{\eta}^{(i)}_{l,k}$ the update equations are:

$$\boldsymbol{x}^{(i)}_{l,k+1} = \boldsymbol{x}_{\text{e},k} \tag{4.34}$$

$$\boldsymbol{\eta}^{(i)}_{l,k+1} = R\big((1 - \tau_k)\boldsymbol{\eta}^{(i)}_{l,k}; \theta, \boldsymbol{k}\big) \,. \tag{4.35}$$

The process is repeated as long as the residual gradient goes below a certain tolerance level.

With the mesh description and retraction operation in place, we proceed to the PPE-MO technique adapted to discrete geometry. The method is broken down into two steps: (a) linearized predictor step, and (b) discrete mesh retraction. The linearized predictor step produces a linear update of the interaction points. In order to obtain a gradient descent-like update, we first consider the first-order update based on the change in UE position. Provided the Hessian is non-singular, from (4.16), the gradient step is obtained as:

$$\Delta \boldsymbol{s}_l = -(\nabla_{\boldsymbol{s}_l} \boldsymbol{u}_l)^{-1} \nabla_{\boldsymbol{x}^{\text{UE}}} \boldsymbol{u}_l \Delta \boldsymbol{x}^{\text{UE}} \,. \tag{4.36}$$

The linear update in (4.36) represents the tangent vector to the solution curve with all interaction points concatenated. For each individual interaction point, the gradient descent step $\boldsymbol{\eta}^{(i)}_l$ is obtained from $\Delta \boldsymbol{s}_l$. However, since the gradient descent update may not remain within the current triangular face if the step size is large, we perform retraction over the triangular mesh to find the updated interaction points. We note that the updated interaction point may fall beyond the boundaries of the neighboring triangular faces. Therefore, the updates are iterated over multiple steps till $\boldsymbol{\eta}_{\text{res}}$ diminishes to zero. A complete listing of the algorithm is provided in Algorithm 2.

The complexity of the proposed manifold-based optimization algorithm for discrete DT is fundamentally independent of the global mesh resolution. The linear predictor step, which

Algorithm 2 Propagation Path Estimation in Discrete Manifold-based DT

Inputs: Discrete mesh $\mathcal{M} = (\mathcal{V}, \mathcal{E}, \mathcal{F})$, Current UE $\boldsymbol{x}^{\mathrm{UE}}$, Next UE $\check{\boldsymbol{x}}^{\mathrm{UE}}$, Path $\boldsymbol{s}_l$
Output: Updated optimal path $\boldsymbol{s}_l^\star$

Phase 1: Linearized Predictor Step
Compute $\Delta \boldsymbol{x}^{\mathrm{UE}} = \check{\boldsymbol{x}}^{\mathrm{UE}} - \boldsymbol{x}^{\mathrm{UE}}$
Compute Hessians: $\nabla_{\boldsymbol{s}_l} \boldsymbol{u}_l$ and $\nabla_{\boldsymbol{x}^{\mathrm{UE}}} \boldsymbol{u}_l$
Update path:

$$\Delta \boldsymbol{s}_l = -(\nabla_{\boldsymbol{s}_l} \boldsymbol{u}_l)^{-1} \left(\nabla_{\boldsymbol{x}^{\mathrm{UE}}} \boldsymbol{u}_l \, \Delta \boldsymbol{x}^{\mathrm{UE}} \right)$$

Extract gradient steps $\{\boldsymbol{\eta}_l^{(1)}, \boldsymbol{\eta}_l^{(2)}, \cdots, \boldsymbol{\eta}_l^{(N_l)}\}$ for each $\boldsymbol{x}_l^{(i)} \in \boldsymbol{s}_l$

Phase 2: Discrete Mesh Retraction
for each interaction point $\boldsymbol{x}_l^{(i)} \in \boldsymbol{s}_l$ with $\boldsymbol{\eta}_l^{(i)}$ **do**
    $\boldsymbol{x}_{l,1}^{(i)} \leftarrow \boldsymbol{x}_l^{(i)}$, $\boldsymbol{\eta}_{l,1}^{(i)} \leftarrow \boldsymbol{\eta}_l^{(i)}$, and $k = 1$
    while $\|\boldsymbol{\eta}_{l,k}^{(i)}\|_2 >$ tolerance **do**
        $\boldsymbol{x}_{\mathrm{tgt}} \leftarrow \boldsymbol{x}_{l,k}^{(i)} + \boldsymbol{\eta}_{l,k}^{(i)}$
        Compute barycentric coordinates of $\boldsymbol{x}_{\mathrm{tgt}}$
        if all barycentric coordinates ≥ 0 **then**
            $\boldsymbol{x}_{l,k+1}^{(i)} \leftarrow \boldsymbol{x}_{\mathrm{tgt}}$
            $\boldsymbol{\eta}_{l,k+1}^{(i)} \leftarrow \boldsymbol{0}$
        else
            Compute intersection point $\boldsymbol{x}_e$ and residual $\boldsymbol{\eta}_{\mathrm{res}}$
            Compute dihedral angle θ and unit vector $\boldsymbol{k}$
            Rotate $\boldsymbol{\eta}_{\mathrm{res}}$ to obtain $\tilde{\boldsymbol{\eta}}_{\mathrm{res}} \leftarrow R(\boldsymbol{\eta}_{\mathrm{res}}; \theta, \boldsymbol{k})$
            $\boldsymbol{x}_{l,k+1}^{(i)} \leftarrow \boldsymbol{x}_e$
            $\boldsymbol{\eta}_{l,k+1}^{(i)} \leftarrow \tilde{\boldsymbol{\eta}}_{\mathrm{res}}$
            $\iota_{\mathrm{curr}} \leftarrow \iota_{\mathrm{next}}$, and $k = k + 1$
        end if
    end while
end for

updates the path based on the UE movement, requires solving a sparse, block-banded linear system which operates in $\mathcal{O}(K)$ time. Furthermore, the retraction phase relies strictly on local barycentric interpolation and sparse incidence matrix lookups to evaluate each geometric boundary crossing in $\mathcal{O}(1)$ operations. Consequently, we have the following remark:

Remark 2. *The computational complexity of PPE-MO in discrete manifold-based DT scales linearly with the local path geometry, yielding a complexity of* $\mathcal{O}(K + E_{\mathrm{cross}})$ *where* E_{cross} *denotes the number of edges intersected during the discrete unfolding process.*

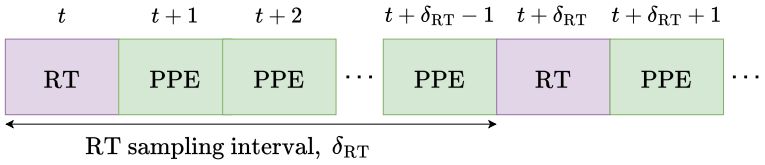


Figure 4.3: A timing diagram describing the RT sampling and PPE-based channel prediction method.

4.3 Numerical Results

This section presents the numerical results of the proposed PPE methods and compares them with conventional schemes through two case studies conducted in Sionna using different scene configurations. Case Study I involves a curved surface around which a UAV flies, emulating the fast movement of a UE. Case Study II considers an indoor scenario in which an autonomous vehicle emulates a slowly moving UE. These studies aim to quantify the impact of surface curvature on channel estimation, as well as the RT sampling frequency required for accurate channel reconstruction. The RT solutions are sampled at different time intervals for use in the proposed PPE method. We denote the RT sampling interval by δ_{RT}. Fig. 4.3 illustrates the sampling scheme of the proposed method. In both case studies, we consider an upper mid-band system operating at 10 GHz, in which the BS and UE are equipped with one omnidirectional antenna each, i.e., $M^{\mathrm{BS}} = M^{\mathrm{UE}} = 1$. The two performance metrics considered are: (1) the channel NMSE, defined as:

$$e_{\mathrm{NMS}} = \frac{\left\| \boldsymbol{H}_{\hat{\mathcal{C}}}(f) - \boldsymbol{H}_{\mathcal{C}}(f) \right\|_{\mathrm{F}}^2}{\left\| \boldsymbol{H}_{\mathcal{C}}(f) \right\|_{\mathrm{F}}^2} \tag{4.37}$$

and (2) the RMSE in the interaction point estimation, defined as:

$$e_{\mathrm{RMS}} = \sqrt{\frac{1}{N_l} \sum_{i=1}^{N_l} \left\| \hat{\boldsymbol{x}}_l^{(i)} - \boldsymbol{x}_l^{(i)} \right\|_2^2}. \tag{4.38}$$

Both e_{NMS} and e_{RMS} performance of the proposed PPE schemes is compared against two benchmark methods: 1) IM-based channel prediction scheme [8]; and 2) MPT-based channel prediction scheme [20]. Note that the IM-based channel prediction scheme requires planar surfaces, for which we utilize the instantaneous planes associated with each interaction point. For MPT, we utilize these instantaneous planes to formulate the objective function.

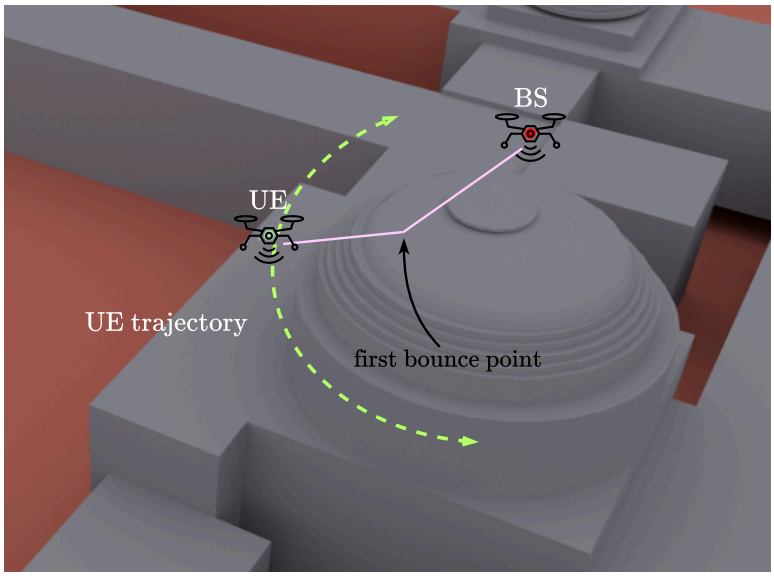


Figure 4.4: Simulation setup for Case Study I. Two UAVs communicate, one acting as a BS and another acting as a UE. The first bounce path is on the curved dome surface.

4.3.1 Case Study I: Curved Surface

The first case study considers a dome-shaped building in which two UAVs, one acting as the BS and the other as the UE, fly near a curved surface. The PPE methods are used to track the interaction points on the dome surface. The simulation setup is illustrated in Fig. 4.4. We utilize a map of the Great Dome at MIT imported from OpenStreetMap. The BS is located at $\boldsymbol{x}^{\mathrm{BS}} = [\,-15.0\,\mathrm{m}\ 45.0\,\mathrm{m}\ 60.0\,\mathrm{m}\,]$. The UE follows a circular trajectory around the dome at a constant angular velocity of $\omega = 2\,\mathrm{degree/s}$ while maintaining a constant radius of $r = 24\,\mathrm{m}$ from the BS. The dome surface is assumed to be metallic, with relative permittivity $\eta = 1$. In this study, we analyze the propagation path reflected from the dome surface. Since the surface is curved, the channel estimation methods must track the interaction point on the surface as the UE position changes in order to maintain estimation accuracy.

Fig. 4.5 plots the NMSE performance of the proposed PPE methods for different sampling intervals δ_{RT}. We observe that the proposed perturbation-based PPE achieves lower NMSE than the manifold-based PPE in the low-sampling-interval regime. This is because, for very small movements, the interaction points undergo only minor changes. As a result, the curved surfaces on which these interaction points lie can be locally approximated as Euclidean, allowing the PPE-PRT to provide an accurate approximation. However, as the distance between $\boldsymbol{x}^{\mathrm{UE}}$ and $\check{\boldsymbol{x}}^{\mathrm{UE}}$ increases, this local Euclidean approximation no longer holds. In such scenarios, the discrete manifold-based PPE outperforms the perturbation-based approach. By traversing the triangular mesh structure, the discrete manifold-based PPE is able to identify interaction points that more closely approximate the true solution. Consequently, for larger sampling intervals, manifold optimization–based PPE provides superior performance compared with PPE-PRT. Additionally, we compare the NMSE performance with conventional schemes, namely IM and MPT. We observe that the NMSE of the IM scheme remains close to that of PPE-PRT. This is because both methods assume planar surfaces; therefore, as the interaction

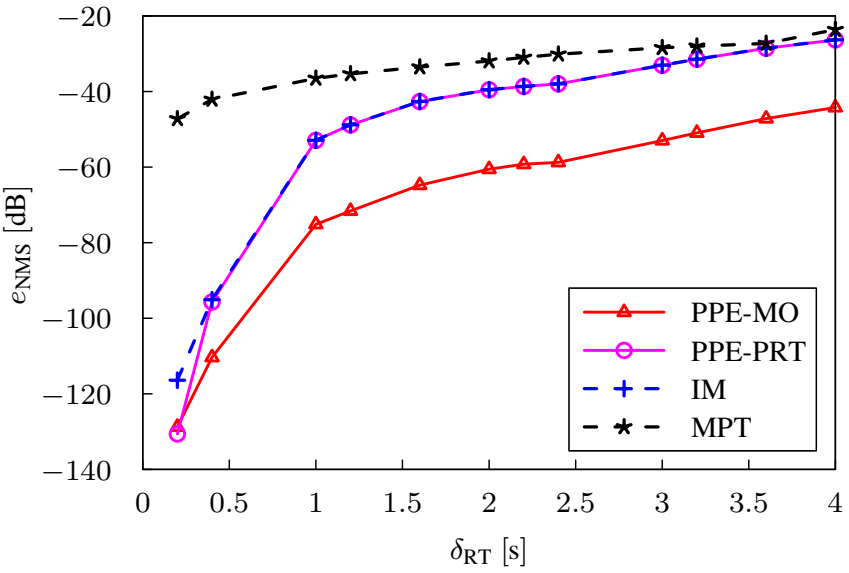


Figure 4.5: A plot showing the NMSE performance e_{NMS} in Case Study I. The RT sampling interval δ_{RT} is varied to control the frequency of obtaining solutions directly from RT.

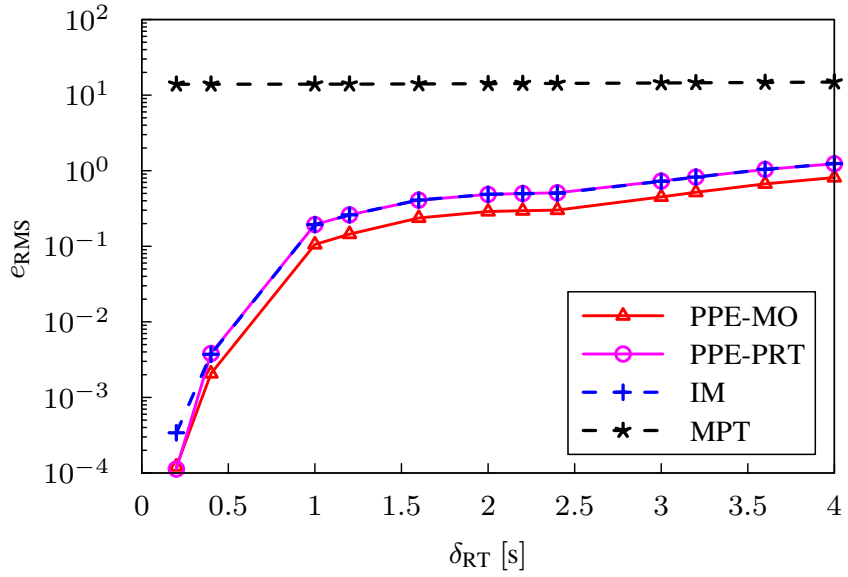


Figure 4.6: A plot showing the RMSE performance e_{RMS} in Case Study I.

point moves along a curved surface, they fail to accurately track its evolution. The MPT method exhibits higher NMSE across all sampling intervals. This performance degradation arises because the underlying optimization problem often yields infeasible solutions. In such cases, MPT reverts to its initial warm start, limiting its ability to adapt to changes in the interaction point. Furthermore, the RMSE performance in Fig. 4.6 shows a similar trend.

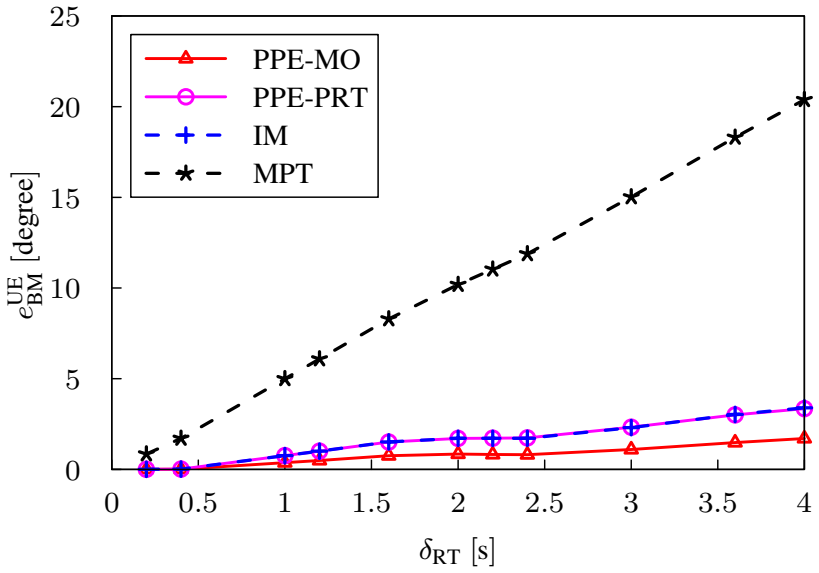


Figure 4.7: A plot showing the beam misalignment error at UE, $e_{\mathrm{BM}}^{\mathrm{UE}}$, for Case Study I.

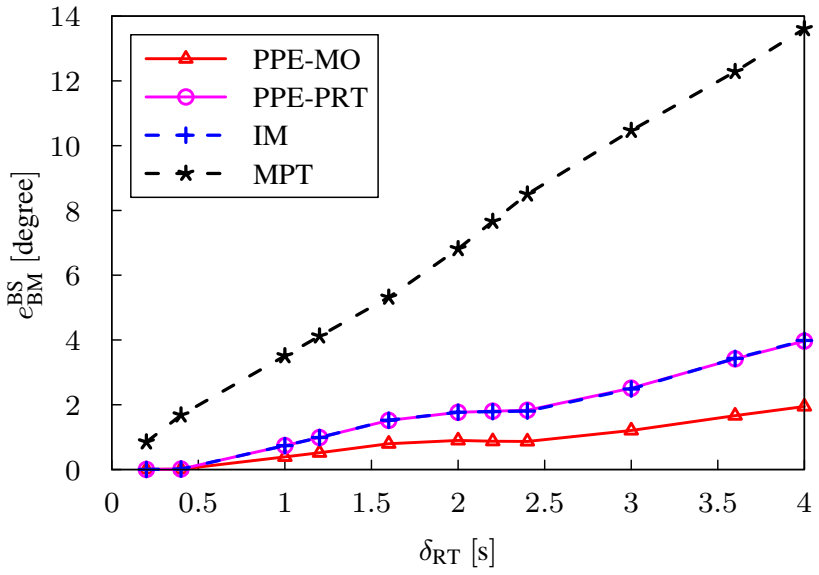


Figure 4.8: A plot showing the beam misalignment error at BS, $e_{\mathrm{BM}}^{\mathrm{BS}}$ for Case Study I.

The estimated spatial coordinates of the interaction points allow the BS to efficiently manage the beam by dynamically orienting toward the strongest signal. In this process, the beam misalignment error is governed by the geometric accuracy of the predicted paths. We quantify the beam misalignment error by the angle between the true direction vectors and the estimated direction vectors of the strongest path. Let $l^{\star}$ denote the strongest propagation

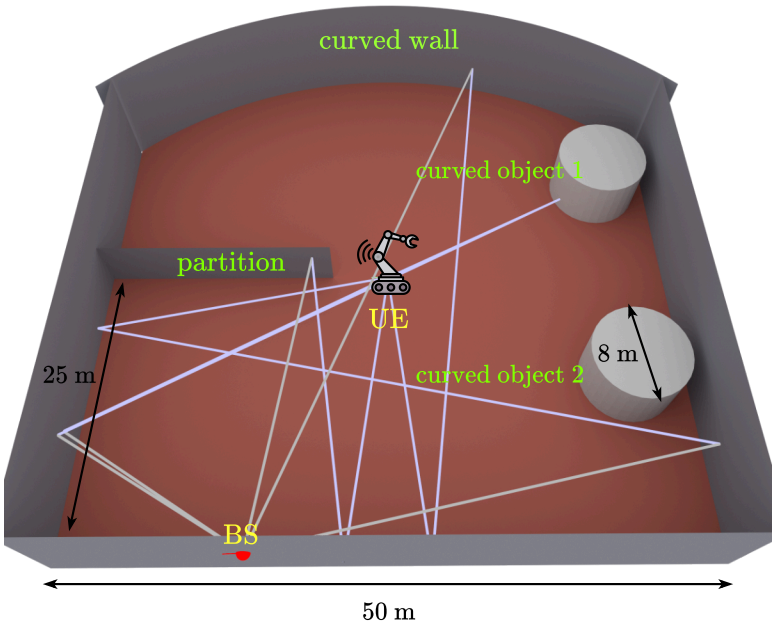


Figure 4.9: Simulation setup for Case Study II. The UE follows a straight trajectory near the BS.

path. The BS and UE beam misalignment errors $e_{\mathrm{BM}}^{\mathrm{BS}}$ and $e_{\mathrm{BM}}^{\mathrm{UE}}$ are defined as

$$e_{\mathrm{BM}}^{\mathrm{BS}} = \arccos\left((\boldsymbol{k}_{l^\star}^{\mathrm{BS}})^{\mathrm{T}}\hat{\boldsymbol{k}}_{l^\star}^{\mathrm{BS}}\right) \tag{4.39}$$

$$e_{\mathrm{BM}}^{\mathrm{UE}} = \arccos\left((\boldsymbol{k}_{l^\star}^{\mathrm{UE}})^{\mathrm{T}}\hat{\boldsymbol{k}}_{l^\star}^{\mathrm{UE}}\right). \tag{4.40}$$

Here, $\hat{\boldsymbol{k}}_{l^\star}^{\mathrm{BS}}$ and $\hat{\boldsymbol{k}}_{l^\star}^{\mathrm{UE}}$ denote the direction vectors of BS and UE obtained from the estimated interaction points in the $l^\star$th path. Fig. 4.7 and 4.8 show the estimation errors for AOA and AOD. The observed performance trends are consistent with those of the NMSE results for the proposed PPE-PRT and PPE-MO methods. In the case of PPE-MO, the angle errors remain within 1.5°, whereas the PPE-PRT technique incurs errors exceeding 3°. Achieving low angle errors is particularly important at higher frequencies, where beams become more directional and sensitive to misalignment. A comparison with conventional schemes shows that MPT fails to provide accurate estimates of AOA and AOD, with angle errors often exceeding 5°. The performance of IM remains close to that of PPE-PRT throughout the range, highlighting the close relationship between these two approaches.

4.3.2 Case Study II: Indoor Environment

The second case study considers an indoor environment, as shown in Fig. 4.9. The scene is created in Sionna with dimensions of 50 m × 50 m. The curved wall has a curvature of 45° along the z-axis. In addition, two cylindrical objects and a partition are included to create a complex multipath scenario, as illustrated in Fig. 4.9. The BS is located at $\boldsymbol{x}^{\mathrm{BS}} = [\,0\,\mathrm{m}\ 50\,\mathrm{m}\ 10\,\mathrm{m}\,]$. The UE, modeling an autonomous vehicle, follows a straight-line trajectory originating from $[\,8\,\mathrm{m}\ -10\,\mathrm{m}\ 0\,\mathrm{m}\,]$ with a velocity of 0.8 m/s.

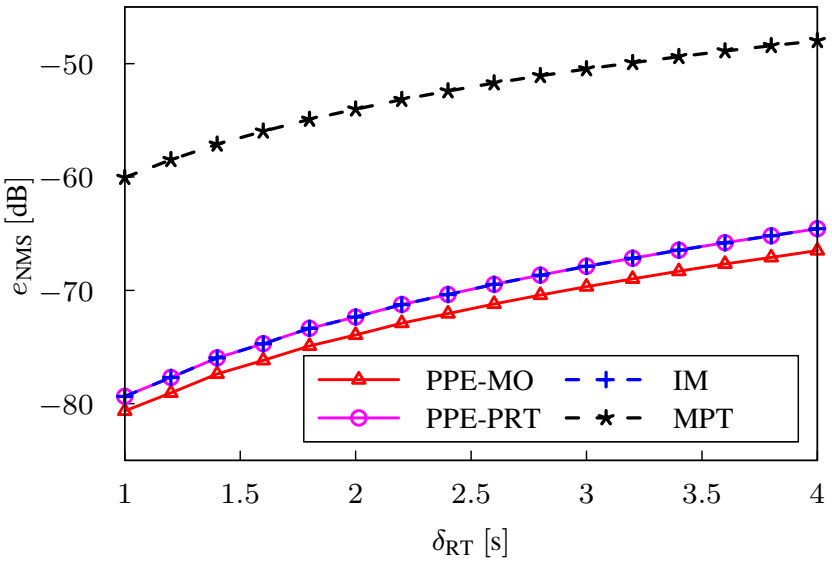


Figure 4.10: A plot showing the NMSE performance of the proposed PPE methods with conventional schemes for Case Study II.

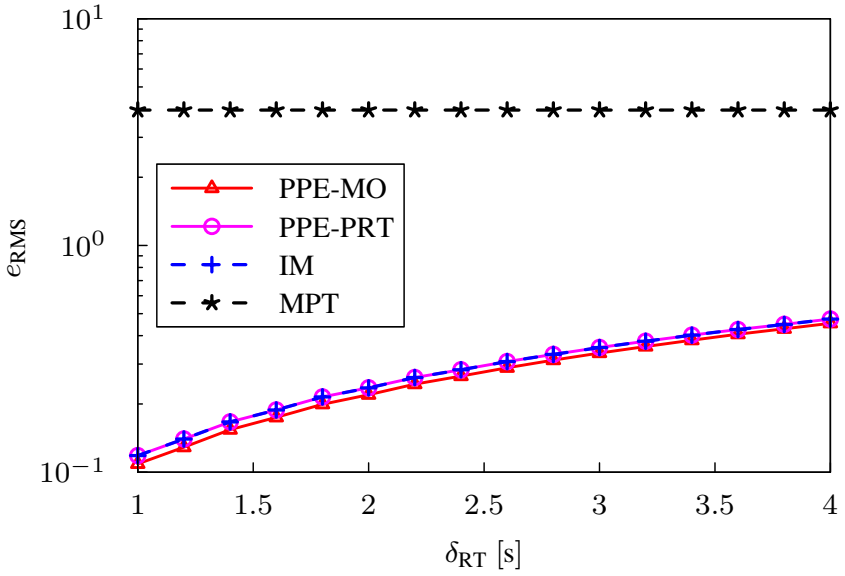


Figure 4.11: A plot comparing the RMSE performance of the proposed PPE methods with conventional schemes for Case Study II.

Fig. 4.10 shows the NMSE performance of the proposed PPE schemes in the indoor setting. PPE-MO consistently outperforms PPE-PRT across different displacement levels. This improved performance stems from PPE-MO's ability to track UE movements over larger distances by leveraging the scene geometry. We also observe that the NMSE performance

of IM remains close to that of PPE-PRT. This behavior is mainly due to their reliance on planar approximations of the constraint surfaces, although the two methods use different approaches to compute the new interaction points. In contrast, MPT consistently fails to find an optimal solution and is therefore unable to reliably estimate the interaction points, resulting in a large NMSE. In addition, we evaluate the RMSE performance of the proposed schemes. Fig. 4.11 presents the RMSE performance of the proposed PPE schemes in the indoor setting. Similar trends are observed: PPE-MO consistently outperforms PPE-PRT and the benchmark schemes.

Chapter 5

Efficient Ray Direction Sampling at the Base Station

The computational efficiency of a wireless DT depends critically on the strategy used to sample ray directions when estimating the channel between a BS and a UE using RT. Unlike conventional rendering, where the goal is to estimate radiance over an entire camera field of view, a wireless DT only needs to accurately resolve the propagation paths that contribute to the channel. Consequently, only a small subset of the angular domain contributes significantly to the channel estimate, while the majority of sampled directions carry negligible energy.

As the UE moves, the dominant propagation paths typically evolve smoothly: for displacements smaller than the channel decorrelation distance, the angular distribution of significant paths remains strongly correlated across consecutive time instants. A sampling distribution that is effective at the current UE location therefore remains informative for nearby locations, which suggests reusing information from previous channel estimates.

This chapter exploits that temporal consistency to build an *adaptive, defensive* importance-sampling strategy. Rather than treating each channel estimate independently, the proposal distribution is progressively refined as the UE traverses the environment. Over time, it concentrates on directions that are likely to contribute to the channel, while retaining enough exploration to discover newly emerging paths. The goal is to reduce the variance of the Monte-Carlo estimator, and hence improve channel-estimation accuracy, under a fixed ray-tracing budget.

5.1 Problem Formulation

5.1.1 Ray-Traced Channel from the Base-Station Perspective

We consider the baseband-equivalent, narrowband channel seen from the BS.[1] Let $(\phi^{\mathrm{BS}}, \theta^{\mathrm{BS}})$ denote the angle of departure (AOD) at the BS antenna. The downlink channel at UE position $\boldsymbol{x}^{\mathrm{UE}}$ is

$$h(\boldsymbol{x}^{\mathrm{UE}}) = \int_{\mathbb{S}^2} a(\phi^{\mathrm{BS}}, \theta^{\mathrm{BS}}; \boldsymbol{x}^{\mathrm{UE}})\, e^{-j2\pi f\, \tau(\phi^{\mathrm{BS}}, \theta^{\mathrm{BS}}; \boldsymbol{x}^{\mathrm{UE}})}\, \sin\theta^{\mathrm{BS}}\, \mathrm{d}\theta^{\mathrm{BS}}\, \mathrm{d}\phi^{\mathrm{BS}} \tag{5.1}$$

[1] The wideband case follows by treating each subcarrier independently.

where a and τ are the complex path gain and propagation delay in direction $(\phi^{\mathrm{BS}}, \theta^{\mathrm{BS}})$, f is the carrier frequency, and $\sin\theta^{\mathrm{BS}}\,\mathrm{d}\theta^{\mathrm{BS}}\,\mathrm{d}\phi^{\mathrm{BS}}$ is the solid-angle element. Both a and τ depend on the UE position, so they are parametrized by $\boldsymbol{x}^{\mathrm{UE}}$. For brevity we write the integrand as $g(\phi^{\mathrm{BS}}, \theta^{\mathrm{BS}}; \boldsymbol{x}^{\mathrm{UE}}) = a\, e^{-j2\pi f\tau}$ and drop the explicit $\boldsymbol{x}^{\mathrm{UE}}$ where it is unambiguous.

5.1.2 Monte-Carlo Baseline and the Sparsity of the Integrand

The integral in (5.1) can be estimated by Monte-Carlo integration. Drawing N directions from a reference density $p(\phi^{\mathrm{BS}}, \theta^{\mathrm{BS}})$ gives the unbiased estimator

$$\hat{h}_{\mathrm{MC}}(\boldsymbol{x}^{\mathrm{UE}}) = \frac{1}{N}\sum_{i=1}^{N} \frac{g(\phi_i^{\mathrm{BS}}, \theta_i^{\mathrm{BS}}; \boldsymbol{x}^{\mathrm{UE}})}{p(\phi_i^{\mathrm{BS}}, \theta_i^{\mathrm{BS}})}\,. \tag{5.2}$$

Conventional schemes take p uniform on the sphere, which allocates equal effort to every direction. In realistic environments, however, the channel energy is concentrated in a small angular region corresponding to the valid propagation paths (Fig. 5.1): the integrand is *sparse* on $\mathbb{S}^2$, taking significant values only over a small fraction of directions.

Because few uniformly drawn samples fall in the significant region, most of the budget is spent evaluating negligible terms, yielding a high-variance, slowly converging estimator. Crucially, each sample requires evaluating the path amplitude and delay through an expensive ray-tracing query, so simply increasing N to reduce variance directly increases computational cost. As the scatterers occupy a smaller solid angle, the fraction of useful samples shrinks further. An effective sampler must therefore concentrate samples where the integrand is large while retaining probability everywhere the integrand may be nonzero. This motivates importance sampling.

5.1.3 Importance Sampling and the Optimal Proposal

Importance sampling (IS) is a classical variance-reduction technique. Instead of sampling from p, we draw N directions from a proposal $q(\phi^{\mathrm{BS}}, \theta^{\mathrm{BS}})$ and reweight,

$$\hat{h}_{\mathrm{IS}}(\boldsymbol{x}^{\mathrm{UE}}; q) = \frac{1}{N}\sum_{i=1}^{N} \frac{g(\phi_i^{\mathrm{BS}}, \theta_i^{\mathrm{BS}}; \boldsymbol{x}^{\mathrm{UE}})}{q(\phi_i^{\mathrm{BS}}, \theta_i^{\mathrm{BS}})}, \qquad (\phi_i^{\mathrm{BS}}, \theta_i^{\mathrm{BS}}) \sim q. \tag{5.3}$$

The estimator is unbiased provided $q > 0$ wherever $g \neq 0$. Its variance is minimized by the proposal

$$q^\star(\phi^{\mathrm{BS}}, \theta^{\mathrm{BS}}) \;\propto\; \left|g(\phi^{\mathrm{BS}}, \theta^{\mathrm{BS}}; \boldsymbol{x}^{\mathrm{UE}})\right| = \left|a(\phi^{\mathrm{BS}}, \theta^{\mathrm{BS}}; \boldsymbol{x}^{\mathrm{UE}})\right| \tag{5.4}$$

i.e., the proposal should be proportional to the *path-gain magnitude*, not its power.[2] Since $q^\star$ depends on the unknown gain field, it cannot be used directly. The remainder of this chapter constructs a tractable, online approximation of $q^\star$ from past observations.

[2]For a real, single-signed integrand (5.4) gives a zero-variance estimator; for a complex integrand it minimizes the variance of the modulus. This directly motivates the amplitude weighting used in Section 5.2.2.

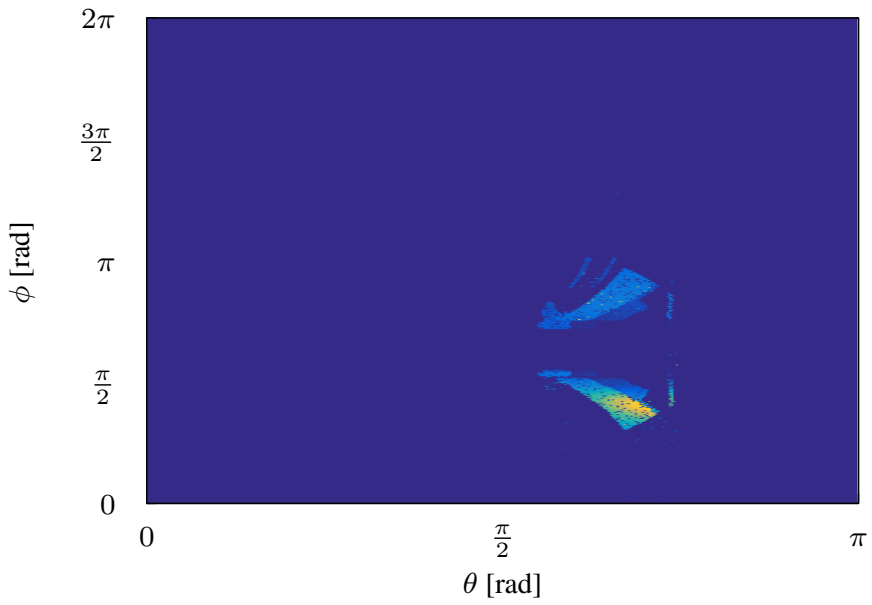


Figure 5.1: Sparsity of the channel over the BS angular domain. Paths with non-negligible energy are confined to $\phi^{\mathrm{BS}} \in [1, 3]$ and $\theta^{\mathrm{BS}} \in [1.5, 2.5]$.

5.1.4 Online Finite-Horizon Formulation

A key difficulty in sampling useful ray directions is that the propagation environment changes with $\boldsymbol{x}^{\mathrm{UE}}$. In practice, the proposal at time t is built from channel information observed at $\tau < t$. Such a proposal tracks persistent paths well but may under-sample directions it has not yet seen: newly emerging clusters (*cluster birth*) receive little probability and are discovered late, while clusters that vanish (*cluster death*) continue to attract samples, wasting budget and inflating the variance.

Consider UE locations $\{\boldsymbol{x}_0^{\mathrm{UE}}, \boldsymbol{x}_1^{\mathrm{UE}}, \ldots, \boldsymbol{x}_T^{\mathrm{UE}}\}$ over a horizon T. At each instant, a proposal q_t is used to estimate the channel with a fixed budget N_{sample}. Let $\mathcal{S}_t$ be the samples at t, $|\mathcal{S}_t| = N_{\mathrm{sample}}$. We seek a *causal* sequence of proposals that minimizes the accumulated mean-squared error,

$$\underset{\{q_t\}_{t=0}^{T}}{\text{minimize}} \quad \sum_{t=0}^{T} \mathbb{E}_{q_t}\Big\{ \|h(\boldsymbol{x}_t^{\mathrm{UE}}) - \hat{h}_{\mathrm{IS}}(\boldsymbol{x}_t^{\mathrm{UE}}; q_t)\|_2^2 \Big\} \tag{5.5}$$

$$\text{such that} \quad |\mathcal{S}_t| = N_{\mathrm{sample}}, \quad q_t = q_t\big(\mathcal{S}_0, \mathcal{S}_1, \ldots, \mathcal{S}_{t-1}\big), \quad \forall t. \tag{5.6}$$

Because $\hat{h}_{\mathrm{IS}}$ is unbiased, the per-step objective equals the estimator variance; minimizing (5.5) therefore amounts to driving each q_t towards $q^\star$ in (5.4) using only past samples. Two features make (5.5) hard: the objective is stochastic and non-convex in q_t, and the target $q^\star$ drifts with the environment, forcing a continual trade-off between *exploiting* observed high-energy directions and *exploring* for new paths.

5.2 Defensive, Adaptive Importance Sampling

5.2.1 Defensive Mixture Proposal

Importance-sampling performance hinges on proposal quality: a proposal over-concentrated on previously observed paths fails to discover births and keeps sampling dead directions, producing a mismatched, high-variance estimator. mitigates this by mixing the learned proposal with a heavy-tailed exploration component of full support [24], [25],

$$q_{\mathrm{DIS},t}(\phi,\theta) = (1-\epsilon_t)\, q_{\mathrm{IS},t}(\phi,\theta) + \epsilon_t\, q_{\mathrm{def}}(\phi,\theta) \tag{5.7}$$

where $q_{\mathrm{IS},t}$ is the learned proposal, q_{def} (e.g. uniform on $\mathbb{S}^2$) guarantees nonzero probability in every direction, and $\epsilon_t \in [0,1]$ controls exploration. Because $q_{\mathrm{def}} > 0$ everywhere, the importance weights $1/q_{\mathrm{DIS},t}$ are bounded on the compact domain, which guarantees a finite-variance estimator regardless of how poorly $q_{\mathrm{IS},t}$ is matched to $q^\star$. The defensive component ensures cluster births can always be detected, while the learned component retains the variance reduction of IS.

Adaptive exploration weight. Rather than fixing ϵ_t, we adapt it to the observed proposal mismatch using the effective sample size,

$$\mathrm{ESS}_t = \frac{\left(\sum_i w_i\right)^2}{\sum_i w_i^2}, \qquad w_i = \frac{1}{q_{\mathrm{DIS},t}(\phi_i^{\mathrm{BS}}, \theta_i^{\mathrm{BS}})}\,. \tag{5.8}$$

A low ESS_t signals a mismatched proposal, in which case ϵ_t is increased to promote exploration; a high ESS_t allows ϵ_t to decay toward a small floor $\epsilon_{\min}$.

5.2.2 Adaptive vMF-Mixture Proposal

Dominant paths in wireless channels cluster in angle because of reflections from common scatterers, and these clusters evolve smoothly for sub-decorrelation displacements. We therefore construct $q_{\mathrm{IS},t}$ from the successful rays of the previous estimate and model it as a mixture of von Mises-Fisher (vMF) distributions—the spherical analogue of the Gaussian and the natural family for directional data on $\mathbb{S}^2$.

For a unit direction $\mathbf{x} \in \mathbb{S}^2$,

$$q_{\mathrm{IS}}(\mathbf{x}) = \sum_{k=1}^{K} \pi_k\, f_{\mathrm{vMF}}(\mathbf{x}; \boldsymbol{\mu}_k, \kappa_k), \qquad f_{\mathrm{vMF}}(\mathbf{x}; \boldsymbol{\mu}, \kappa) = C_3(\kappa)\, e^{\kappa \boldsymbol{\mu}^\top \mathbf{x}} \tag{5.9}$$

with mixture weights $\pi_k \geq 0$, $\sum_k \pi_k = 1$, mean directions $\boldsymbol{\mu}_k$, concentrations κ_k, and normalizer $C_3(\kappa) = \kappa/(4\pi \sinh \kappa)$.

By (5.4), an accurate approximation of $q^\star$ requires the mixture to track the *gain magnitude* across angle, which motivates the amplitude weighting below.

Weighted histogram. Discretize the angular domain into M bins; bin i has angles (ϕ_i, θ_i), complex gain g_i, and direction

$$\mathbf{x}_i = \begin{bmatrix} \sin\theta_i \cos\phi_i \\ \sin\theta_i \sin\phi_i \\ \cos\theta_i \end{bmatrix} \in \mathbb{S}^2 \,. \tag{5.10}$$

Consistent with (5.4), we weight each bin by its gain *amplitude*,

$$w_i = \frac{|g_i|}{\sum_{j=1}^{M} |g_j|}. \tag{5.11}$$

Gain-weighted EM. The parameters are obtained by maximizing the gain-weighted log-likelihood

$$\mathcal{L} = \sum_{i=1}^{M} w_i \log\Big(\sum_{k=1}^{K} \pi_k f_{\mathrm{vMF}}(\mathbf{x}_i; \boldsymbol{\mu}_k, \kappa_k)\Big) \tag{5.12}$$

via expectation-maximization [26].
E-step:

$$r_{ik} = \frac{\pi_k f_{\mathrm{vMF}}(\mathbf{x}_i; \boldsymbol{\mu}_k, \kappa_k)}{\sum_{l=1}^{K} \pi_l f_{\mathrm{vMF}}(\mathbf{x}_i; \boldsymbol{\mu}_l, \kappa_l)} \,. \tag{5.13}$$

M-step:

$$\pi_k = \frac{\sum_i w_i r_{ik}}{\sum_l \sum_i w_i r_{il}}, \qquad \mathbf{R}_k = \sum_{i=1}^{M} w_i r_{ik} \mathbf{x}_i, \qquad \boldsymbol{\mu}_k = \frac{\mathbf{R}_k}{\|\mathbf{R}_k\|_2}, \tag{5.14}$$

$$\bar{R}_k = \frac{\|\mathbf{R}_k\|_2}{\sum_i w_i r_{ik}}, \qquad \frac{I_{3/2}(\kappa_k)}{I_{1/2}(\kappa_k)} = \bar{R}_k \implies \kappa_k \approx \frac{\bar{R}_k\,(3 - \bar{R}_k^2)}{1 - \bar{R}_k^2} \tag{5.15}$$

where I_ν is the modified Bessel function of the first kind and the closed-form κ_k is the standard $p = 3$ approximation. EM is initialized with spherical k-means and iterated until $\mathcal{L}$ changes by less than a tolerance δ or N_{iter} is reached; components with vanishing π_k or degenerate $\bar{R}_k$ are pruned.

Model order and birth/death. The number of components K is over-provisioned and pruned by the mixture weights, so dead clusters ($\pi_k \to 0$) are removed automatically. Newly born clusters are first discovered by the *defensive* samples of (5.7); because those samples carry gain, they enter the next weighted histogram and are absorbed into the mixture at the following EM update. Thus births are detected within one step of appearing, and ϵ_t's adaptation shortens this latency further.

Sampling and complexity. At the next instant, rays are drawn from (5.7): with probability ϵ_t from q_{def}, otherwise by selecting component k with probability π_k and drawing from $f_{\mathrm{vMF}}(\cdot; \boldsymbol{\mu}_k, \kappa_k)$. Compared with uniform sampling, this allocates most of the budget to directions likely to contribute to the received signal while retaining support over all clusters

Algorithm 3 Gain-Weighted vMF Mixture Estimation from an Angular Histogram

Require: Histogram $\{(\phi_i, \theta_i, g_i)\}_{i=1}^{M}$, components K, max iterations N_{iter}, tolerance δ
Ensure: vMF mixture $\{\pi_k, \boldsymbol{\mu}_k, \kappa_k\}_{k=1}^{K}$

$\mathbf{x}_i \leftarrow [\sin\theta_i \cos\phi_i,\ \sin\theta_i \sin\phi_i,\ \cos\theta_i]^\top$
$w_i \leftarrow |g_i| / \sum_j |g_j|$
Initialize $\{\pi_k, \boldsymbol{\mu}_k, \kappa_k\}$ by spherical k-means
for $n = 1$ to N_{iter} **do**
    E-step: $r_{ik} \leftarrow \dfrac{\pi_k f_{\text{vMF}}(\mathbf{x}_i; \boldsymbol{\mu}_k, \kappa_k)}{\sum_l \pi_l f_{\text{vMF}}(\mathbf{x}_i; \boldsymbol{\mu}_l, \kappa_l)}$
    M-step: update π_k, $\mathbf{R}_k$, $\boldsymbol{\mu}_k$, $\bar{R}_k$, κ_k as above; prune degenerate components
    $\mathcal{L} \leftarrow \sum_i w_i \log\big(\sum_k \pi_k f_{\text{vMF}}(\mathbf{x}_i; \boldsymbol{\mu}_k, \kappa_k)\big)$
    if $|\Delta\mathcal{L}| < \delta$ **then**
        break
    end if
end for
return $\{\pi_k, \boldsymbol{\mu}_k, \kappa_k\}_{k=1}^{K}$

and, through q_{def}, the whole sphere. The EM update costs $O(N_{\text{iter}} MK)$ scalar operations, which is negligible relative to the N_{sample} RT queries it replaces; hence the overhead of adapting the proposal is amortized by the reduction in required rays.

Because $q_{\text{IS},t}$ is refreshed after every estimate, it adapts to the temporal evolution of the channel: persistent clusters are continuously refined, dead clusters decay, and new clusters are absorbed. The sampler thus progressively concentrates effort on the most informative directions while remaining responsive to changes in the environment.

5.3 Implementation of Adaptive vMF-Mixture Proposal

5.3.1 Warping Uniform Samples to the vMF Mixture

The importance sampler must draw ray directions from the proposal

$$q_{\text{IS}}(\mathbf{x}) = \sum_{k=1}^{K} \pi_k f_{\text{vMF}}(\mathbf{x}; \boldsymbol{\mu}_k, \kappa_k) \tag{5.16}$$

of Section 5.2.2. In Sionna, a directional sampler is exposed as a deterministic *warp* that maps a pair of uniform variates to a point on the sphere, so we require a map

$$W : [0,1)^2 \longrightarrow \mathbb{S}^2, \qquad \mathbf{x} = W(\mathbf{u}), \quad \mathbf{u} = (u_1, u_2) \sim \mathcal{U}\big([0,1)^2\big), \tag{5.17}$$

whose pushforward of the uniform measure equals q_{IS}. Since the mixture CDF on $\mathbb{S}^2$ has no closed-form inverse, we realise W as the composition of two exact steps:

1. **Component selection:** draw an index $k \in \{1, \ldots, K\}$ with probability π_k by inverting the discrete CDF (Section 5.3.1).

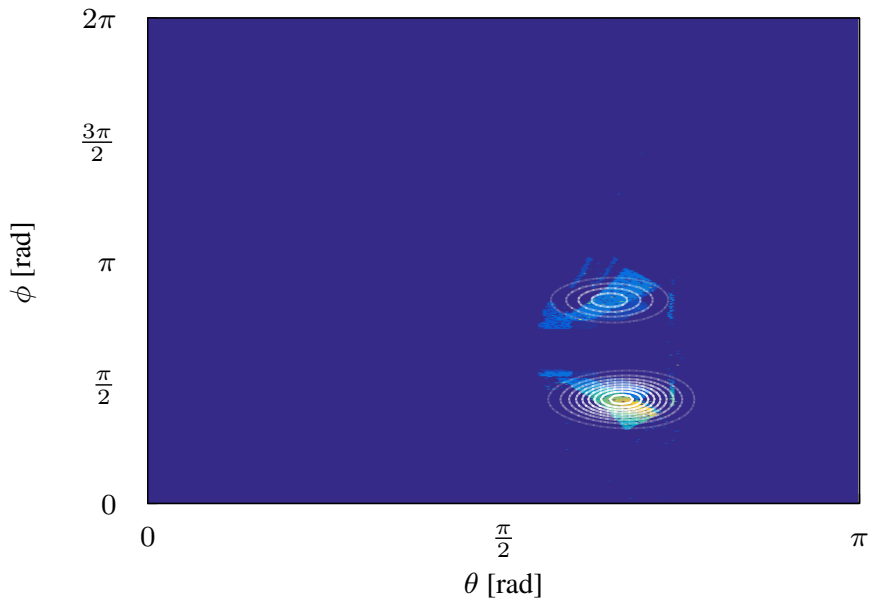


Figure 5.2: Example gain-weighted vMF mixture fitted to the angular histogram.

2. **Conditional vMF sampling:** draw $\mathbf{x} \sim f_{\text{vMF}}(\cdot; \boldsymbol{\mu}_k, \kappa_k)$ by exact inversion of the polar-cosine marginal together with a uniform azimuth (Section 5.3.1).

Because the warp exposes only two uniform numbers, we reuse u_1 for both the component selection and the polar inversion through the remapping of Section 5.3.1, reserving u_2 for the azimuth.

Step 1: Component Selection and Sample Remapping

Let the cumulative mixture weights be $F_0 = 0$ and $F_k = \sum_{j=1}^{k} \pi_j$ for $k = 1, 2, \ldots, K$, so that $F_K = 1$. The selected component is

$$k^\star(u_1) = \min\{\, k : u_1 < F_k \,\} \tag{5.18}$$

i.e. the unique k with $u_1 \in [F_{k-1}, F_k)$; hence $\Pr[k^\star = k] = F_k - F_{k-1} = \pi_k$. To avoid consuming a third uniform, we rescale u_1 within the chosen bin to a fresh variate,

$$\tilde{u}_1 = \frac{u_1 - F_{k^\star - 1}}{\pi_{k^\star}} \in [0, 1). \tag{5.19}$$

Since u_1 is uniform on the sub-interval $[F_{k^\star-1}, F_{k^\star})$ of length $\pi_{k^\star}$, this affine map gives $\tilde{u}_1 \sim \mathcal{U}([0, 1))$.

Step 2: Sampling a Single vMF Component

Fix the selected parameters $(\boldsymbol{\mu}, \kappa) = (\boldsymbol{\mu}_{k^\star}, \kappa_{k^\star})$ and work in a local frame whose z-axis is aligned with $\boldsymbol{\mu}$. A direction is parametrised by the polar cosine $w = \boldsymbol{\mu}^\top \mathbf{x} \in [-1, 1]$ and an azimuth $\varphi \in [0, 2\pi)$.

Polar cosine. Integrating f_{vMF} over the azimuth gives the marginal of w,

$$g(w) = \frac{\kappa}{2\sinh\kappa}\, e^{\kappa w}, \qquad w \in [-1, 1], \qquad G(w) = \frac{e^{\kappa w} - e^{-\kappa}}{2\sinh\kappa}. \tag{5.20}$$

Setting $G(w) = \tilde{u}_1$ yields the exact inverse

$$w = 1 + \frac{1}{\kappa}\,\log\!\left(\tilde{u}_1 + (1 - \tilde{u}_1)\, e^{-2\kappa}\right). \tag{5.21}$$

For large κ, $e^{-2\kappa} \to 0$ and the expression degrades gracefully to the numerically stable form $w \approx 1 + \kappa^{-1} \log \tilde{u}_1$.

Azimuth and local direction. The azimuth is uniform, $\varphi = 2\pi u_2$ with $u_2 \sim \mathcal{U}([0, 1))$. With $s = \sqrt{1 - w^2}$, the direction in the $\boldsymbol{\mu}$-aligned frame is

$$\mathbf{x}_{\mathrm{loc}} = \left[s\cos\varphi,\ s\sin\varphi,\ w\right]^{\mathrm{T}}. \tag{5.22}$$

Rotation to the global frame. Let $\mathbf{Q}(\boldsymbol{\mu}) \in \mathrm{SO}(3)$ be any rotation mapping the local z-axis $\mathbf{e}_3$ to $\boldsymbol{\mu}$, i.e. $\mathbf{Q}(\boldsymbol{\mu})\,\mathbf{e}_3 = \boldsymbol{\mu}$.

The warped sample is

$$\mathbf{x} = W(\mathbf{u}) = \mathbf{Q}(\boldsymbol{\mu}_{k^\star})\,\mathbf{x}_{\mathrm{loc}}. \tag{5.23}$$

Orthogonality of $\mathbf{Q}$ gives $\|\mathbf{x}\| = \|\mathbf{x}_{\mathrm{loc}}\| = 1$, so $\mathbf{x} \in \mathbb{S}^2$.

Correctness

The warp is exact. Conditional on $k^\star = k$, the pair (w, φ) follows the vMF polar marginal and uniform azimuth, so $\mathbf{x} \sim f_{\mathrm{vMF}}(\cdot; \boldsymbol{\mu}_k, \kappa_k)$. Marginalising over the component selection,

$$p_W(\mathbf{x}) = \sum_{k=1}^{K} \Pr[k^\star = k]\, f_{\mathrm{vMF}}(\mathbf{x}; \boldsymbol{\mu}_k, \kappa_k) = \sum_{k=1}^{K} \pi_k\, f_{\mathrm{vMF}}(\mathbf{x}; \boldsymbol{\mu}_k, \kappa_k) = q_{\mathrm{IS}}(\mathbf{x}), \tag{5.24}$$

i.e., the pushforward density equals the proposal. Drawing from the defensive mixture $q_{\mathrm{DIS},t}$ of (5.7) then only requires selecting q_{def} with probability ϵ_t and applying W otherwise. Reusing u_1 for both component selection and polar inversion introduces a deterministic coupling between $k^\star$ and $\tilde{u}_1$, but the sampled distribution remains exact because $\tilde{u}_1$ is uniform conditionally on each $k^\star$. The only cost is a mild loss of cross-component stratification; an independent third uniform for selection removes the coupling at the expense of the two-argument warp signature.

5.4 Numerical Results

This section evaluates the proposed sampling scheme numerically. We consider a channel operating at a carrier frequency of $f = 10$ GHz, simulated on an OpenStreetMap model of the Great Dome at MIT. The scene is discretized into a 100×100 grid of UE locations,

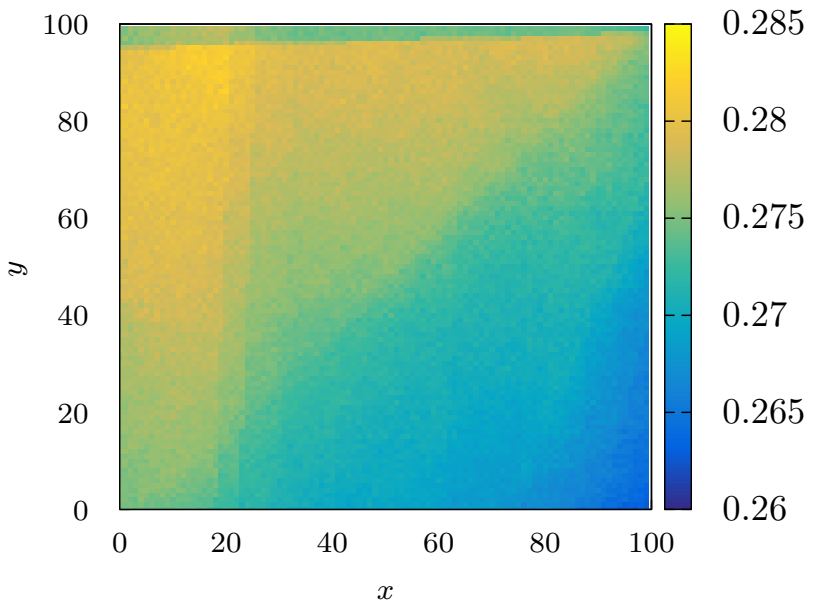


Figure 5.3: Sampling efficiency of the importance sampling-based ray tracer. At each UE location, 10,000 rays are launched from the DIS proposal, and the efficiency is averaged over 100 runs.

each cell measuring 0.8 m $\times$ 0.7 m. We denote the grid coordinates with X and Y, $X, Y \in \{1, 2, \cdots, 100\}$.

We assess performance using the *sampling efficiency*, defined as the ratio of successful rays to launched rays. At each UE location, 10,000 rays are launched, and the efficiency is averaged over 100 independent runs.

We first compare the proposed DIS scheme against uniform sampling for a fixed proposal. Rays are launched from the defensive proposal q_{DIS} with mixing weight $\epsilon = 0.01$, which combines a uniform defensive distribution with the vMF mixture q_{IS}. The mixture is estimated once at grid location $(0, 0)$ using $K = 4$ components and is then reused at all subsequent locations without adaptation.

Figure 5.3 shows the resulting sampling efficiency of DIS. Across all locations, the efficiency exceeds 25%, indicating that roughly one in four launched rays lands on a valid propagation path. In contrast, uniform sampling achieves an efficiency of only about 1.2% (Fig. 5.4), so DIS improves the sampling efficiency by more than an order of magnitude.

We next evaluate the sampling efficiency in an adaptive setting, in which the UE follows a diagonal trajectory across the grid. Here the proposal q_{IS} is re-estimated from the successful rays of the previous location as the UE moves, so the mixture continuously tracks the evolving propagation environment. Figure 5.5 compares the sampling efficiency of DIS and uniform sampling along this trajectory.

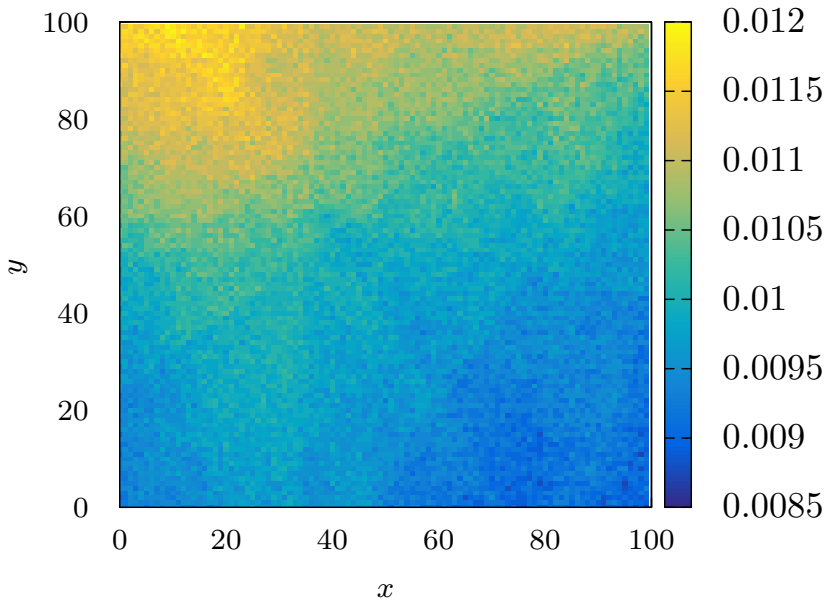


Figure 5.4: Sampling efficiency of the uniform-sampling-based ray tracer. At each UE location, 10,000 rays are launched from the uniform distribution, and the efficiency is averaged over 100 runs.

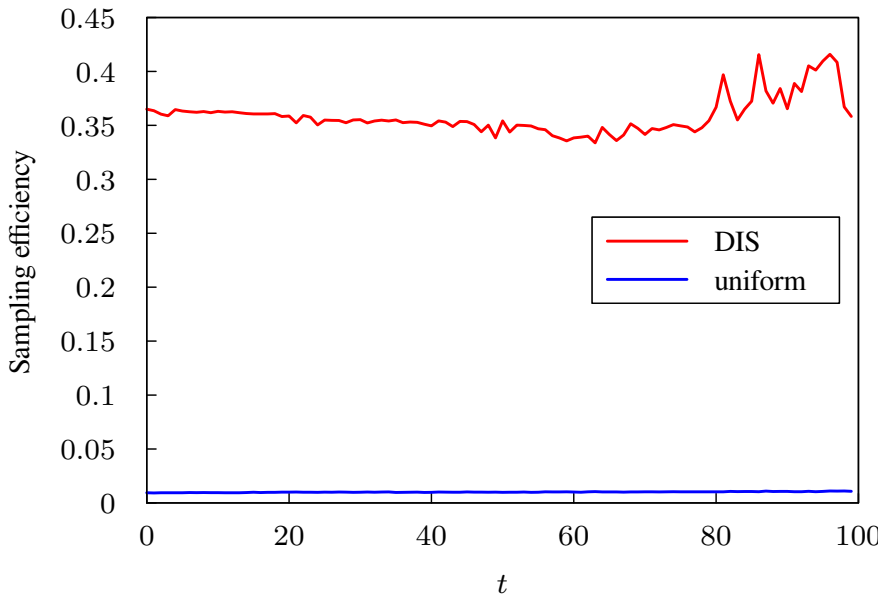


Figure 5.5: Sampling efficiency of DIS compared with the uniform-sampling-based ray tracer along a diagonal UE trajectory. At each UE location, 10,000 rays are launched from the corresponding distribution, and the efficiency is averaged over 100 runs.

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